COMPLETED NATIVE CAMPAIGN · NEW RESULTS / FROZEN HISTORICAL APPENDIX

Fair rho tuning and the prepared IC tail

New optimization addendum · exact retained degree-83 a=1 model · run-20261012T0528Z

Frozen selection: Fold backend · 64 lockstep lanes · DP6 · escape off · step-cap factor 2000.

156 rho solves and 5 prepared IC target queries independently replayed; 0 invalid answers.

Five disjoint held-out target blocks under the declared conditional orbit-exclusion law.

Selected rho has lower held-out process cost than default rho: native default/selected geometric ratio 1.7567 [1.5402,2.0334]. Prepared IC has four shorter cases and one 13,034.488 ms tail. Its geometric IC/rho ratio 0.4310 [0.1380,1.7764] and arithmetic mean-time ratio 1.5426 point in opposite directions. There is no definitive overall online winner. The combined edition retains the prior 48 pages as a historically frozen appendix.

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SECTION 01

What the fairer rho control changes

> The registered search selected fold-L64-DP6-E0. On 5 held-out target blocks, default/selected rho process cost is 1.756671 [1.540157,2.033353]. Prepared IC’s geometric ratio is 0.431011 [0.137988,1.776388], including parity; its arithmetic mean-time ratio is 1.542600 in the opposite direction. No overall definitive IC/rho online winner is established.

Rho was tuned over a frozen two-stage field-kernel and policy search, rather than left at its original default. The selected policy was fixed before disjoint-target holdout. Its held-out mean process costs are default 3.915205 s versus selected 2.152616 s. This improves the tested control within the registered scope; it is not a global optimum or a claim that the cold-selected policy is an online optimum.

Prepared IC is shorter than the selected-rho three-seed mean on four target cases, but the first target takes 13,034.487993 ms. All five cases are retained. A geometric mean summarizes multiplicative target ratios; the arithmetic mean describes this panel’s average time and is sensitive to the tail. Their disagreement and the interval crossing 1 prevent a blanket online-winner conclusion.

The old packed-reference 8.1813× observation used a weaker control and is inadmissible as evidence against strong rho. Its correction and the prior default-strong evaluation remain historical. The appended 48-page report is unchanged. New selected-rho times are never paired with old IC times, and the historical 218.034569 cold ratio is not revised.

Native target-block estimator	Geometric ratio	95% interval	Arithmetic mean-time ratio
Default/selected rho process	1.756671	[1.540157, 2.033353]	1.818813
Default/selected rho online	1.836263	[1.609766, 2.135352]	1.904154
Prepared IC/selected rho online	0.431011	[0.137988, 1.776388]	1.542600

Sources: O01, O04, O05, O06.

SECTION 02

Declared corpus, sampling law and evidence flow

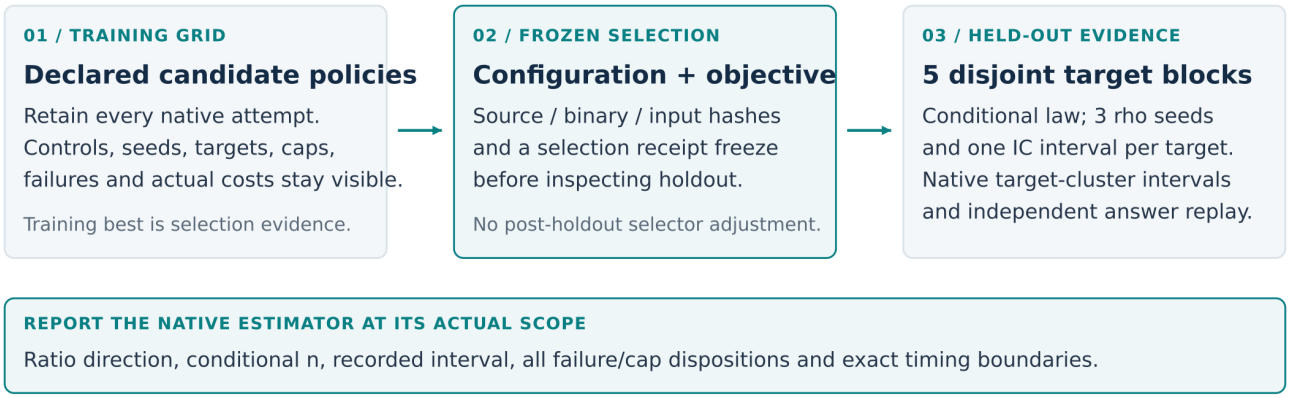
The exact model is `icv1-f2m83-t6151469093347-9eba6cd5`: $y^2+xy=x^3+x^2+1$ over GF(2)[z]/(z^83+z^7+z^4+z²+1), subgroup r=8569786107849059, cofactor 1128547018, G=(661424465139818486029201 5,886826041002924939153952 9). Field degree 83 is distinct from the approximately 52.928-bit subgroup. No substitution of the [0,1,2,45] representation or a=0 case occurs.

Native generation uses unbiased nonzero scalar proposals, rejecting signed-Frobenius-orbit overlap with the old exposed point, retained K600 base and other generated points. Three training and five held-out target blocks are sampled without replacement under these exclusions. They are not unconditional independent draws. The validation scalar sidecar is not solver input.

Kernel and policy training use nine cases on three targets; holdout uses five new targets and three fresh rho seeds per target/arm. There is one prepared IC interval per target and one shared IC preparation. No fastest seed, per-target policy choice or post-holdout tuning is used.

A selection boundary before the held-out result

One exact field/subgroup/point binding; actual configuration and outcome come from native receipts.



126 training rho processes precede the frozen selection; 30 held-out rho solves and 5 prepared IC target queries follow.
Prior default-policy pages remain historically frozen; this addendum supplies separately source-bound new evidence.

Executed campaign flow: 126 training rho processes; one frozen policy; 30 held-out rho solves plus 5 prepared IC target intervals. Statistical units remain five target blocks. [O01–O05]

Sources: O01, O02, O03, O05, O06.

SECTION 03

Native training grid and the frozen selection

Kernel training compares Wide and Fold at 32 lanes/DP4/escape off on 3 targets × 3 seeds: 18 launches. All 18 verify. In nine matched cases, recovered answers, walk counters and reported charges agree. No full step-by-step trajectory transcript is exported. The native selector freezes Fold under the registered whole-process criterion.

Kernel / policy	Verified cases	Mean process s	Mean online ms
fold-L32-DP4-E0	9/9	3.800763	3594.509619
wide-L32-DP4-E0	9/9	4.905513	4700.309334

Fold then receives all 12 policies: lanes 16/32/64 × DP4/6 × escape off/on, nine cases each. All 108 verify uncapped. Eligibility requires all nine verified uncapped cases; wrong answers reject. The selection score is native mean whole-process wall, with the default preferred within 2% of best. This preference is a selection rule, not a confidence test.

Fold policy	Verified cases	Mean process s	Mean online ms
fold-L16-DP4-E0	9/9	4.254725	4051.613049
fold-L16-DP4-E1	9/9	4.318931	4116.507651
fold-L16-DP6-E0	9/9	2.699743	2492.910388
fold-L16-DP6-E1	9/9	2.737213	2532.174321
fold-L32-DP4-E0	9/9	3.695433	3490.948152
fold-L32-DP4-E1	9/9	3.746029	3540.834438
fold-L32-DP6-E0	9/9	2.341132	2133.465168
fold-L32-DP6-E1	9/9	2.355200	2150.567076
fold-L64-DP4-E0	9/9	3.473313	3268.471999
fold-L64-DP4-E1	9/9	3.498884	3293.263616
fold-L64-DP6-E0	9/9	2.146530	1946.016033
fold-L64-DP6-E1	9/9	2.172581	1964.917797

The native choice is Fold/64/DP6/off, step-cap factor 2000: training process mean 2.146530315777778 s and secondary online mean 1946.0160328888887 ms. It is the selected fastest eligible tested policy under the registered cold-training criterion. The staged search does not test all 24 backend-policy combinations; kernel-policy interactions, other lane/DP values, jump tables, normal-coordinate point storage and compiler exploration remain outside scope.

Sources: O01, O02, O03, O07.

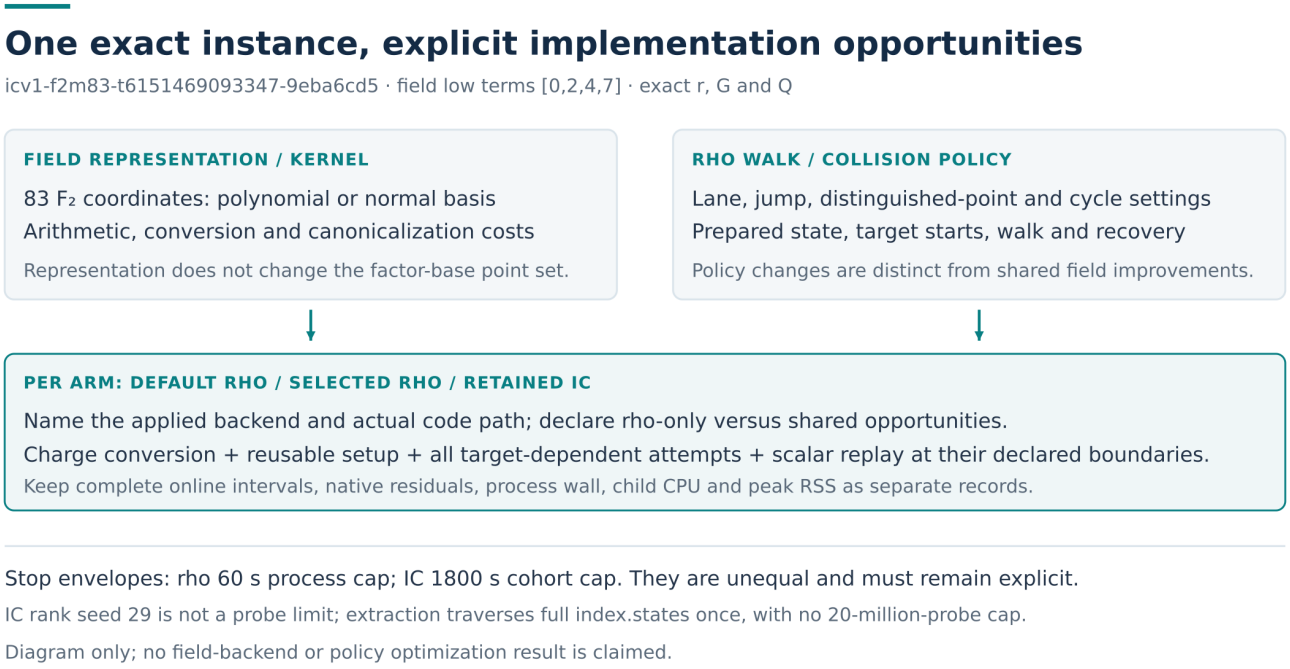
SECTION 04

Field primitives shared fairly with the strong algorithm

Wide and Fold run the same generic strong walk, canonicalization, collision recovery and scalar transport in one release executable. Fold delegates field arithmetic to the existing Gf2_128 primitives already available to IC: carryless tail-fold reduction and specialized squaring. This is field engineering on the same polynomial, subgroup and point law; it does not change the factor-base set or introduce a new vector-space basis.

Both kernels use runtime hardware dispatch and the same default release/compiler profile, without asymmetric target-cpu/LTO/PGO flags. Strong canonicalization already uses its exact zero-run/least-rotation selector; an older full-rotation-scan profile is not current optimization evidence.

Nine same-input kernel cases match verified outcomes, walk counters and reported charges. Field tests and the unchanged generic algorithm support the comparison, but neither a full step transcript nor identical primitive-field/instruction counts is established. Changing lane, DP and escape settings can change useful collision progress; ns/step alone did not select the policy.



Exact-instance and backend/policy fairness. The selected policy shares existing IC arithmetic opportunities while preserving the strong group algorithm. Rho 60 s and IC 1800 s process caps remain unequal. [O01,O07]

Sources: O01, O02, O07, O08.

SECTION 05

Selected rho on held-out target blocks

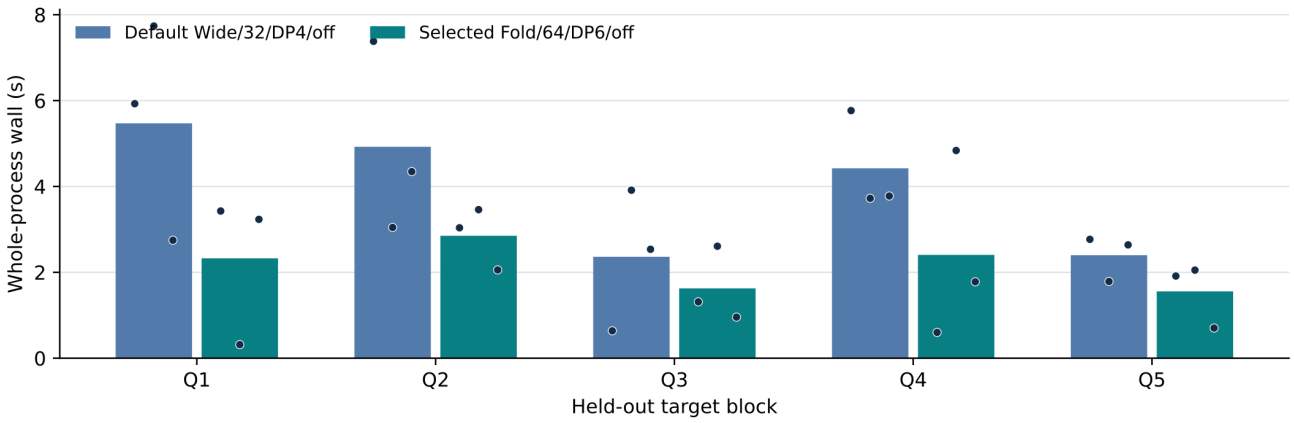
All 30 held-out rho processes complete and verify; no target or seed is dropped. The selected policy and original Wide/32/DP4/off default each receive three fresh seeds per target under the same rho 60 s process cap. Default/selected whole-process ratio 1.756671 has native 95% interval [1.540157,2.033353]; its arithmetic mean-time ratio 1.818813 is a separate descriptive value.

The native primary averages three actual rho process times within each target/arm, then takes the geometric mean of five target ratios. It bootstraps complete target blocks 20,000 times with seed 202610128399. The interval is conditional on the frozen selection, orbit-exclusion target law, these five blocks and this host; resampling is an exchangeability approximation, not fifteen independent targets.

Target	Default process mean s	Selected process mean s	Native default/selected
holdout_1	5.470036	2.325667	2.352029
holdout_2	4.923624	2.850779	1.727116
holdout_3	2.361908	1.625409	1.453117
holdout_4	4.422436	2.405613	1.838382
holdout_5	2.398020	1.555612	1.541529

Held-out rho: all 30 seed solves and native target means

Bars are native three-seed means; dots are all recorded individual process times.



Five target blocks under the orbit-exclusion law; no caps or failed answers. The policy was frozen before holdout.

Source: native statistics.json · run-20261012T0528Z

All 30 held-out rho seed process times (dots) and native three-seed target means (bars), run-20261012T0528Z. All answers independently replayed. [O04,O05]

Sources: O01, O04, O05.

SECTION 06

Prepared IC: all five cases and opposite estimator directions

> IC/selected-rho geometric mean 0.431011 [0.137988,1.776388] includes parity. IC arithmetic mean 3008.682028 ms versus rho 1950.396640 ms yields native ratio 1.542600 in the opposite direction. Keep both summaries and the 13,034.487993 ms case; neither establishes a definitive overall online winner.

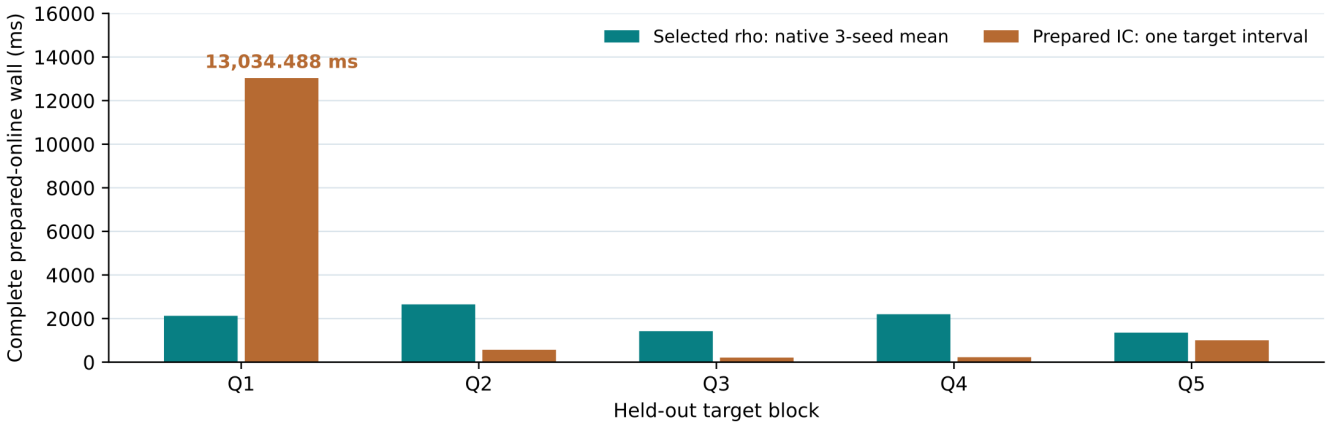
IC has one measured target interval per held-out point, versus three selected-rho seed draws averaged per point. It is lower on four of five target cases. Q1’s verified 13,034.487993 ms interval stays in every estimator and the displayed panel. It is not deleted or replaced as an inconvenient outlier.

The native geometric target-ratio estimator uses 20,000 target-block bootstrap draws, seed 202610128400. The wide interval reflects a small five-target panel under the conditional generation law. It does not measure repeated IC setup, repeated IC timings on each point, host variation, or target populations beyond that law.

Target	Selected rho native mean ms	Prepared IC ms	Native IC/rho
holdout_1	2125.592165	13034.487993	6.132168
holdout_2	2650.855295	567.676432	0.214148
holdout_3	1421.803807	209.946099	0.147662
holdout_4	2199.462871	227.662621	0.103508
holdout_5	1354.269060	1003.636996	0.741091

All five IC cases: four short intervals and one tail

No case is removed. IC is prepared once; each rho target mean includes three fresh seed draws.



Geometric IC/rho ratio 0.4310 [0.1380,1.7764]; arithmetic mean-time ratio 1.5426 points the other way.

Native estimators; no definitive overall online winner. Source: statistics.json · run-20261012T0528Z

All five complete prepared IC intervals against selected-rho native three-seed means. Four shorter cases and one long tail give opposite geometric/arithmetic summaries; geometric CI includes 1. [O04]

Sources: O01, O04, O05, O06.

SECTION 07

What is charged: warm queries and the single IC cohort

The immutable original IC binary pays once for loading the retained base, root indexing and guided rank/log preparation, then solves five targets from shared immutable prepared state. Recorded preparation is 1149151.699878 ms. The entire five-target process wall is 1164.439319887 s, CPU 1164.339530 s and wait4 peak RSS 5811478528 bytes. These are one cohort receipt, not five cold single-target measurements.

Original construction/selection of the 600 representatives and OS-cold page-cache flushing are unmeasured. Do not divide preparation by five and call the result a cold single-target cost. Do not pair new rho times with old IC times; the old 218.034569 five-pair retained-input cold ratio stays frozen.

Rho’s process includes launch/load, curve/normal basis, G-only jumps, Q validation, walk/collision, generic replay and receipt output. Dataset generation and 126 training solves are explicit research costs, not charged anew to a selected production solve. Stock IC assumes prevalidated fixtures whereas rho charges Q validation online; this asymmetry remains visible.

Rho process cap is 60 s; IC cohort cap is 1800 s. IC’s third CLI argument 29 is a rank seed, not a 20-million-probe cap; extraction traverses full index.states once. All selected-rho seeds and five IC targets completed uncensored, so the declared complete-online comparison is available, with the unequal caps still disclosed.

The new outer monitor polls at 5 ms, with nominal 0–5 ms exit-observation delay plus scheduling/poll-body effects—not a strict bound. This differs from the old 100 ms observer. Native complete online clocks remain separate from outer wall and child CPU. Raw target rows preserve the phase sum; no phase sum is substituted for complete online.

Native receipt	Recorded quantity
IC reusable preparation (one seed 29)	1149151.699878 ms
IC whole 5-target cohort wall	1164.439319887 s
IC cohort child CPU	1164.339530 s
IC cohort peak RSS	5811478528 bytes
Observed process monitor	5 ms polling; no strict scheduling bound
Common calibrated work S	null; IC probes and rho steps remain different units

Sources: O01, O04, O06, O08.

SECTION 08

Independent verification, limits and measurement decisions

Standalone native C++ replay reports PASS: 156 rho answers plus 5 IC answers independently verified, all 5 actual IC four-point witnesses checked and 0 invalid rows. It reproduces fixture generation/orbit exclusions and verifies 99,600 retained base point/label records. No full 600-row rank/factor-log transcript or step-by-step rho walk transcript is supplied.

Four native isolation records—kernel, policy, holdout and IC—report exit 0 and 0 contended samples. The host is Linux x86_64 with exposed AMD EPYC 9V74 model, one pinned CPU 3 and 12 GiB address-space limit. These guest observations do not certify physical-host exclusivity, second-host reproduction or sealed ecbench admission.

Recorded pre-timing validation passes 4 worker tests, 20 native runner/statistics tests and the default release library suite 3947 passed / 0 failed / 112 ignored. Conductor was unavailable. The exact legacy case still lacks the applicable sealed wide IC adapter, calibrated common S, independent-host gates and complete rank/log evidence; no admitted ledger promotion or global optimum is claimed.

Measured source commits 4c9da3bd/4ae7caa2 and immutable executable hashes remain the execution identity. The delivery tree later integrates main at 010e4f69; this does not retime or relabel measured binaries. Pre-measurement checks above belong to their captured stage. Separate post-merge delivery checks pass: release library 3992 passed / 0 failed / 112 ignored in 402.32 s, and the configured example 4 passed / 0 failed in 12.19 s. These checks do not replace the pre-measurement results or measured-source identities.

For future measurement, freeze the objective before selection: cold process versus prepared-online are different policies. Report both geometric ratios and expected-time/descriptive arithmetic summaries, retain tail cases, and expand the target panel with explicit sampling law and repeated setup/host measurements. This panel changes the reference-fairness evidence while leaving the overall prepared-online ranking unresolved.

Native authoritative artifact	SHA-256
Statistics	6250c42f5a7f6644db5c7c39bd5c712155b2667d081309b5e99202f64f8918f7
Independent final replay	6640a9f69e8ef66f3e31a3eab70c18925bdc8140a623aac3559c01fdeb57e294
Machine protocol	bdbed17abe2c9073089d52498d481e42fe67df6b6ae1953467b20125c1faa9db
Executed runner	b1b94b24fea42a7ac02e2ce21dfd913e7ba4a2b66a536f48d65bb3cfe8044a27
Executed rho worker	21827dcc645fd4591b39dd913619f047d267a98d541854bd7c83ee7143d8bbcb
Immutable IC worker	983f28400b7852fc6628ba97d3ea83b2ab9e32d9944658c7eea831ee89a4abf2

Sources: O01, O04, O05, O06, O07, O08, O11.

SECTION 09

Source register and frozen historical appendix

This addendum is a source-bound optimization and accounting report from recorded native results. Python only formats those records and renders plots/pages; it estimates no ratio, mean, interval, selection score or arithmetic certificate. Every kernel/policy case and every held-out rho/IC row remains linked below in native records.

The combined PDF places this addendum before the unchanged historical 48 pages. Their default-policy conclusions retain their original scope and dates. The historical packed-reference correction, exact-field identity erratum and old 218.034569 cold ratio remain provenance, not newly optimized measurements. Historical pages are preserved during merging.

ID	Durable native source
O01	Frozen human protocol: PROTOCOL.md
O02	Kernel selection and all 18 native cases: records/panel/kernel/selection.json
O03	Policy selection and all 108 native cases: records/panel/policy/selection.json
O04	Authoritative native statistics: records/panel/statistics.json
O05	Independent final arithmetic replay: verification/final_receipt.json
O06	Measurement/estimand review: MEASUREMENT_REVIEW.md
O07	Field/policy implementation scope audit: OPTIMIZATION_SCOPE_AUDIT.md
O08	Executed IC cohort provenance: records/panel/ic/stage_manifest.json
O09	Raw held-out rho rows: records/panel/holdout/records.jsonl
O10	Native runner/statistics source: native_runner/src/statistics.rs
O11	Campaign delivery, checks and graph disposition: README.md

Figures: native recorded training scores, all 30 held-out rho seed values, all 5 IC intervals and native target-block estimates. Report identity: run-20261012T0528Z. Layout/merge QA sits beside this source.

RUN-20261012T0226Z · ACCOUNTING AND REFERENCE CORRECTION

Degree-83 IC against the strong rho reference

Exact retained a=1 / K600 fixture · completed five-pair native diagnostic

One public target; five paired seed rounds and five same-seed A/A control pairs.

All 22 children completed and independently replayed; no process failures or caps.

Source baseline: a8194d616ed8c04489ec46177d369f1b1670360a.

> Observed IC/strong cost: 1.3823× prepared online and 218.0346× fresh-process wall from retained base. These are native paired geometric means on one Q and one host. The historical 8.1813× packed-reference observation does not establish a strong-reference advantage.

Contents

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SECTION 01

Diagnostic conclusion and the historical correction

> For this fixed point and frozen five-pair panel, the native IC/strong paired geometric mean is 1.3823 [1.0358, 1.8823] online and 218.0346 [172.3584, 285.5737] for fresh-process wall. Values above 1 mean greater IC cost. No controlled vs_rho row is promoted.

The source-bound comparison does not reproduce the previous synthesis’s interpretation of a degree-83 online lead against the required reference. Its 8.1813× median recorded rho/IC ratio used the old packed reference. That worker stores every step, scans the orbit in polynomial coordinates and inverts per step. Current reference rules exclude it as the primary strong control. The old scalar answers remain valid; its reference interpretation is corrected.

The fresh reference uses distinguished points, normal-basis canonicalization and batched inversion over 32 lockstep lanes in one CPU worker. Both fresh methods solve the exact same public Q without receiving its scalar. The standalone replay verifies every one of the 22 outcomes, including the two excluded warm-ups.

The intervals describe five seed/repeat pairs on this one Q and one host. They do not estimate performance on independently drawn targets, other hosts or deployed curves. The whole-process comparison begins with the retained base file and omits original base selection/construction; it is not a construction-from-curve total.

Recorded estimator	Value	95% native interval / n
IC/strong prepared-online cost	1.382330	[1.035774, 1.882300] · 5 pairs
IC/strong fresh-process wall	218.034569	[172.358384, 285.573727] · 5 pairs
Strong A / strong B online control	0.997036	[0.967928, 1.023496] · 5 pairs
Archived packed/IC median	8.181302	Historical only; different reference and estimator

Sources: R02, R03, R04, R07.

SECTION 02

Exact field, subgroup, point and frozen design

The correct exact-field slug is `icv1-f2m83-t6151469093347-9eba6cd5`. Full identity: `ICV1:f2m-83-a5fd0d39:6151469093347:967140655691088192855606 2:0x1:-7:unk:r:9eba6cd5d359`. The frozen protocol and original archive replay copied the wrong cdcc5432 alias, which belongs to low terms [0,1,2,45]. Their actual [0,2,4,7] field and G/Q arithmetic are unchanged. The separate erratum preserves frozen evidence.

Field degree 83 is distinct from the subgroup's approximately 52.928 bits. This is the retained a=1 instance, not the a=0 degree-83 / 81-bit ECC2K confidence gate. Exact polynomial, subgroup and G/Q coordinates are the primary binding.

The panel requests two excluded warm-ups, five same-seed strong/strong A/A pairs, and five interleaved strong/IC pairs; all 22 children completed verified. Strong seeds are 202610061–202610065 and IC rank seeds 21–25. Matched order alternates by round. One fixed Q is repeated in every child.

Item	Frozen value
Curve and field	$y^2+xy=x^3+x^2+1$; GF(2)[z]/(z^83+z^7+z^4+z²+1)
Prime subgroup r / cofactor h	8569786107849059 / 1128547018
Generator G	(661424465139818486029 2015,886826041002924939153952 9)
Public Q	(265589834553716618053296 0,388152630799167326529261 5)
Retained base	600 signed-Frobenius representatives; 99,600 usable points
IC workers	Frozen compact-orbit producer; rank threads 1, target threads 1
Strong policy	Wide-field exact-point adapter; 32 lockstep lanes; DP bits 4; step-cap factor 2000; fruitless escape false
Envelope	CPU 3, one pinned CPU; 12 GiB RLIMIT_AS; 1800 s process-group timeout

Sources: R01, R05, R06, R07.

SECTION 03

Three spaces and a one-dimensional subgroup

The arithmetic field has 83 coordinates over F2. The retained base is a point-defined union of 600 signed-Frobenius orbits, not a small linear subspace of those field coordinates. Its relation labels are formal vectors in (Z/rZ)^600. The cyclic prime-order subgroup itself has one scalar dimension over Z/rZ; 600 columns do not create 600 independent group directions.

A normal basis changes field representation and makes Frobenius squaring a rotation; it does not change factor-base membership. The displayed four-vector witness is retained, not newly discovered. Independent replay checks the actual four-point sum and scalar answer; the full 600-row preparation transcript remains missing.

One exact pipeline, three distinct spaces

Field degree: 83 · prime-subgroup size: approximately 53 bits · r = 8569786107849059

The cyclic subgroup has one scalar dimension over Z/rZ; the 600 IC coordinates are formal coefficient labels.

01 / FIELD COORDINATES

83 coordinates over F_2

$F = F_2[z] / (z^{83} + z^7 + z^4 + z^2 + 1)$

Polynomial and normal bases represent the same field values.

Squaring rotates normal coordinates. Least rotations define orbit keys.

Not a factor-base membership rule.

encodes →

02 / FACTOR-BASE POINT SET

600 orbit representatives

$B = \{ \varepsilon\, \tau^k(R_j) \}$

$0 \le k < 83$; $j = 0,\ldots,599$; $\varepsilon = \pm 1$

99,600 distinct signed-orbit points

Membership is defined by points. This B is not a linear subspace of the 83 field coordinates.

→ labels

03 / RELATION COEFFICIENTS

Vectors in $(Z/rZ)^{600}$

$v(\varepsilon\, \tau^k(R_j)) = \varepsilon\, \lambda^k\, e_j$

e_j selects one orbit/log column. Coefficients track phase and sign.

These 600 column coordinates are separate from field encoding and the actual point-set B.

SIGNED-FROBENIUS TRANSPORT ON THE PRIME SUBGROUP

$\tau(x,y) = (x^2, y^2) \cdot -P = (x, x+y) \cdot \tau(P) = [\lambda]P$

$\lambda = 2998004823950457 \cdot \lambda^2 - \lambda + 2 = 0 \pmod r$ · exact order 83

RETAINED FOUR-POINT WITNESS: $P_1 + P_2 + P_3 + P_4 = Q$

Point indices [87298, 27061, 36601, 66281] · signs and shifts independently replayed in the archived receipt

$$v_Q = \lambda^{74} e_{525} - \lambda e_{163} - \lambda^{40} e_{220} - \lambda^{23} e_{399}$$

FACTOR-LOG DOT PRODUCT

$L_j = \log_G(R_j) \cdot d = v_Q \cdot L \pmod r$

→

INDEPENDENT FINAL REPLAY

$[d]G = Q$

Full factor-log/rank transcript is not retained: final witness replay does not certify every preparation equation.

`Exact legacy model: icv1-f2m83-t6151469093347-9eba6cd5 · primary binding: explicit field + r + G + Q`

Source: verification/INVARIANTS.md and verification/CURVE_ID_ERRATUM.md · correctness diagram; no runtime claim

Exact retained signed-orbit labels and four-vector witness, from INVARIANTS.md and the independent replay. The diagram describes existing algebra and supplies no runtime claim. [R04–R06]

Sources: R04, R05, R06.

SECTION 04

The invariant shared by IC and strong rho

On the subgroup, τ(P)=[λ]P with λ=2998004823950457, λ²−λ+2=0 mod r and exact order 83. Signed Frobenius multiplies each attached scalar coefficient by ελ^k. IC records that transport in its factor-log columns; rho records it in two state scalars for P=[a]G+[b]Q. Both are formal labels in the same one-dimensional subgroup.

At a rho distinguished-point collision, the nonzero denominator b1−b2 permits d=(a2−a1)/(b1−b2) mod r. A zero denominator yields no recovered scalar. IC instead extracts a four-point witness and applies its factor-log dot product. Each final answer requires [d]G=Q replay. The shared invariant does not make their preparation, target work or runtime equivalent.

The same invariant, two coefficient systems

83 F_2 field coordinates · one scalar dimension in the prime-order subgroup · λ = 2998004823950457

$C = \varepsilon\,\tau^k(P) = [\varepsilon\,\lambda^k]P$ · forward phase transport on the same exact group

IC / 600 FORMAL FACTOR-LOG COLUMNS

$v(P) \in (Z/rZ)^{600}$

$v(C) = \varepsilon\,\lambda^k\,v(P)$

For $P = R_j$: $v(P) = e_j$; $L_j = \log_G(R_j)$.

Root-index extraction yields a four-point witness and its coefficient vector v_Q.

$d = v_Q \cdot L \pmod r$

Full rank/log preparation transcript remains absent.

STRONG RHO / TWO FORMAL STATE SCALARS

$P = [a]G + [b]Q$

$(a,b) \to (\varepsilon\,\lambda^k a,\ \varepsilon\,\lambda^k b)$

Equal canonical collision points imply

$(a_1 - a_2) + (b_1 - b_2)d = 0 \pmod r$.

$d = (a_2 - a_1) / (b_1 - b_2) \pmod r$

Requires $b_1 - b_2 \neq 0$; zero gives no scalar.

Two labels do not mean a two-dimensional subgroup.

Independent final-answer replay: [d]G = Q

Same algebraic transport does not imply the same work or runtime. Source: verification/INVARIANTS.md.
Exact legacy model: icv1-f2m83-t6151469093347-9eba6cd5 · method comparison only; no fresh timing claim

Existing forward signed-Frobenius transport in the retained IC labeling and strong rho state. This is a method comparison, not a fresh identity or timing claim. [R05]

Sources: R05, R07.

SECTION 05

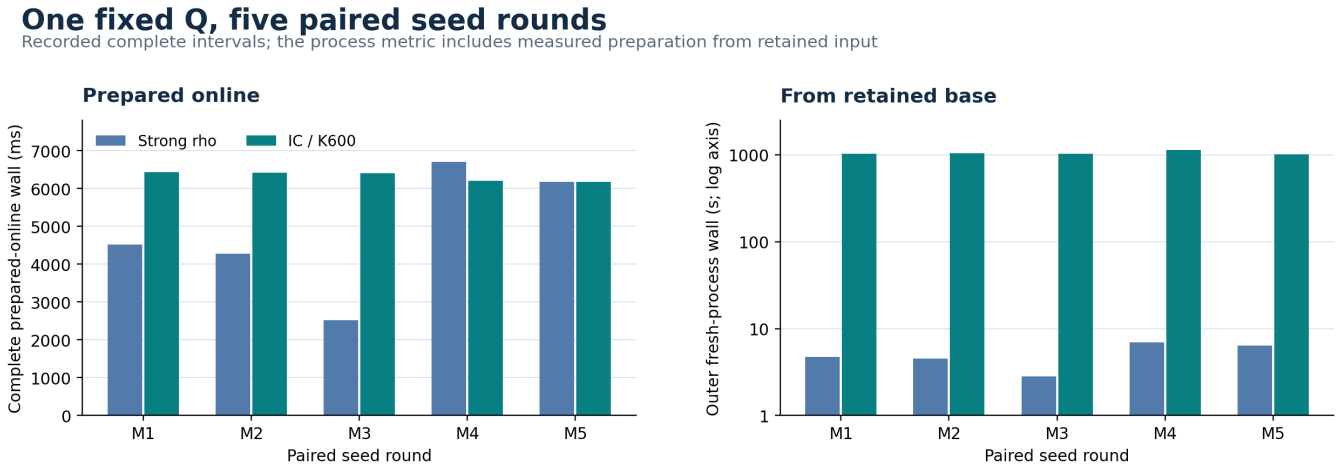
Per-round recorded online and process values

Online values are each producer’s complete declared target interval. The primary statistic does not replace those intervals with a sum of component clocks. Every matched row solves Q; there are no exclusions, failed answers, overall caps or missing matched receipts.

Strong online includes Q membership/subgroup validation, target-dependent starts, walk/collision work and generic scalar replay. The IC target stage assumes the retained fixture is prevalidated; that declared validation asymmetry remains visible.

M4 has a lower measured IC online interval: 6204.636907 ms versus strong 6698.575519 ms, native strong/IC ratio 1.079608. M5 is near a tie, native ratio 0.999600. The conditional paired geometric mean over all five rounds still gives greater IC cost; this does not assert that strong is faster in every seed round.

Pair / order	Strong online ms	IC online ms	Strong/IC native ratio
M1 · rho→IC	4516.159970	6437.368839	0.701554
M2 · IC→rho	4279.021477	6416.446390	0.666883
M3 · rho→IC	2514.613847	6408.147497	0.392409
M4 · IC→rho	6698.575519	6204.636907	1.079608
M5 · rho→IC	6176.824943	6179.295727	0.999600



Recorded intervals for run-20261012T0226Z, n=5 seed pairs on one Q. Right: fresh-process wall from retained base, logarithmic axis; includes measured reusable preparation. [R02,R03]

Sources: R02, R03, R04.

SECTION 06

Native uncertainty and same-seed A/A controls

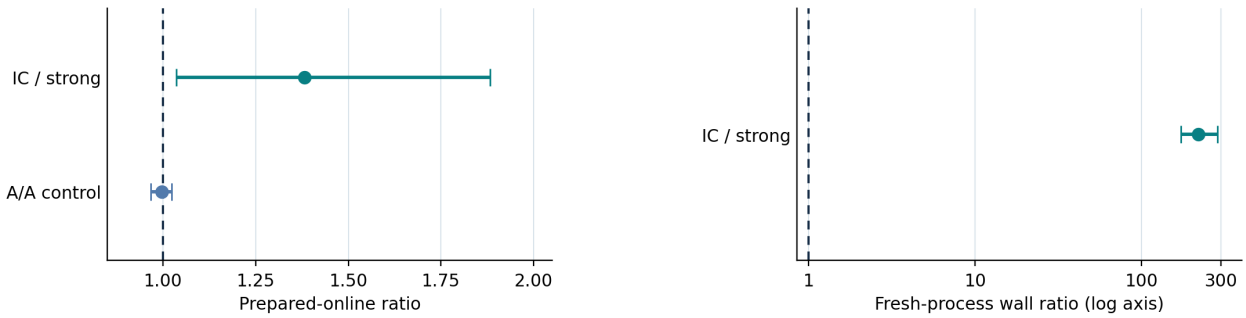
The frozen estimator takes the arithmetic mean of within-pair log(strong/IC) ratios and exponentiates it. It uses 20,000 paired percentile-bootstrap resamples. The runner applies seed 2026101283 to A/A and seed+1=2026101284 to matched pairs. The native secondary exporter presents IC/strong as the reciprocal, with interval [1/upper,1/lower]; it performs no new resampling or estimator change.

These geometric means are not ratios of arithmetic mean times. The A/A control’s paired geometric mean is 0.997036, interval [0.967928,1.023496], with observed online ratios spanning 0.948233–1.026892. It diagnoses repeat noise at fixed seeds on this host; it does not establish physical-host exclusivity.

n=5 is a small conditional panel. Seed sensitivity and repeated measurement of one Q cannot supply a fresh-target extraction-tail distribution. Keep all raw pairs, avoid population-wide significance claims, and prospectively freeze independent unseen targets for the next question.

Native paired geometric means and 95% intervals

IC/strong values are native reciprocals of the frozen strong/IC estimator; no new resampling



IC/strong > 1 means greater IC cost; parity is the dashed line. A/A compares two strong runs with the same seed.
Each interval is conditional on n=5 paired repeats of one Q and one host. It is not a fresh-target population interval.
Sources: statistics.json + native report_data_secondary.json · run-20261012T0226Z

Native paired geometric means and recorded 95% intervals, run-20261012T0226Z. IC/strong >1 means greater IC cost; A/A is strong-A/strong-B. Each interval covers five pairs on one Q and one host. [R02,R03]

Sources: R01, R02, R03, R10.

SECTION 07

Fresh-process cost from the retained base

The outer process starts from the retained factor-base file. IC loads its point arrays, builds the root index and acquires/solves its guided rank/log preparation. It does not newly select/build the original 600 representatives. OS page cache is not flushed. Thus cold means fresh process-local state, not OS-cold or total construction-from-curve runtime. Adding original construction cost would add IC work, but its unmeasured value stays unknown.

Strong curve/normal-basis construction and G-only jumps are measured. Supplied subgroup/catalog facts are parameters, not newly timed curve discovery or factorization. The native means below are secondary descriptive arithmetic means and ranges over five successful formal runs; they replace neither the paired primary estimator nor its interval.

Metric: mean [min,max]	Strong	IC
Preparation ms	191.516 [186.924, 194.744]	1047924.983 [1009528.197, 1139341.566]
Process wall s	5.068 [2.804, 6.909]	1054.552 [1015.960, 1145.812]
Child user+system CPU s	5.029 [2.709, 6.887]	1054.438 [1015.897, 1145.722]
Outer peak RSS bytes	116588544.0 [55263232.0, 208527360.0]	5811525222.4 [5811470336.0, 5811605504.0]

- IC prepared-online extraction is measured after its reusable rank/log preparation. Fresh cold children reconstruct index/log state independently; no cross-process warm cache is inherited.
- The retained root table has 29,878,182 entries. Table-entry counts are stored states, not independently verified factor logs or useful-row counts.
- The root/index and guided preparation must stay charged in this workload. Amortized reuse across new targets requires an explicitly different workload count.

Sources: R03, R05, R07.

SECTION 08

Phase residuals and the outer wall boundary

IC PDP subtracts its nested final-lift check from extraction wall at one target thread. That check is only the final addition/equality; earlier lift work stays under PDP. All five formal IC rows have exact five-phase conformance=false. Strong's three component timers also leave nonzero residuals; its five-phase flag is inapplicable/null. The native primary uses complete online intervals.

Outer wall starts before spawn and ends when wait4(WNOHANG) observes exit. The runner polls every 100 ms: wall includes nominal 0–100 ms detection delay plus scheduling/poll/heartbeat effects, not a strict 100 ms bound. Child CPU excludes parent sleeping. Formal M3 strong reports producer process_before_receipt_io=2.707912963 s, CPU=2.708705 s and parent wall=2.803706954 s. These boundaries do not prove an exact causal overhead decomposition. Frozen wall values remain untouched.

IC retains five component clocks. Each residual below is the native complete interval minus its reported phase sum; no phase sum is silently substituted for online.

Pair	Query ms	PDP ms	Lift ms	Descent ms	Replay ms	Residual ms
M1	0.010927	6437.211479	0.012360	0.000611	0.103506	0.029956
M2	0.000030	6416.329133	0.012258	0.000640	0.103356	0.000973
M3	0.000020	6408.030341	0.012148	0.000601	0.103426	0.000961
M4	0.000020	6204.505849	0.025700	0.000630	0.103586	0.001122
M5	0.000071	6179.177809	0.012648	0.000831	0.103577	0.000791

Strong reports Q query/validation, walk/collision and generic replay. The different component definitions are preserved.

Pair	Q validation ms	Walk/collision ms	Generic replay ms	Residual ms
M1	0.232440	4515.698654	0.228475	0.000401
M2	0.232550	4278.561313	0.227333	0.000281
M3	0.248956	2514.137047	0.227503	0.000341
M4	0.261765	6698.084148	0.229306	0.000300
M5	0.233803	6176.356336	0.234494	0.000310

Both native producers have nonzero timer residuals. The complete-interval primary and descriptive process metrics remain available as diagnostic observations; exact phase-accounting admission remains unmet.

The parent observer-clock caveat is retained: before-spawn→observed-exit includes polling and scheduling/heartbeat effects. CPU and producer process timestamps corroborate it without retroactively adjusting wall. Future harnesses should use event-driven pidfd/SIGCHLD or native blocking wait timing.

Sources: R03, R07, R10.

SECTION 09

Counters and independent correctness at their actual scope

> Independent answer/base/witness correctness is established at the stated replay scope. It does not certify every missing preparation equation, reproducibility of wall time, physical-host isolation or a speedup across independent targets.

IC returns the same four indices [87298,27061,36601,66281] and 8,845,441 target probes in each formal repeat of Q. Probe units count partner candidates reaching canonicalization/lookup, not every S3 call or group addition. The producer records rank=600, 600 admitted rank relations and zero rank failures; the complete rank equations and factor-log vector are not exported.

The standalone C++ replay revalidates the field/curve/subgroup, all 600 representatives and 99,600 point/label records, every actual IC four-point witness and all 22 recovered answers. It reports PASS, 22 independently verified answers and zero invalid rows. The validation-only scalar sidecar is checked after answers, not passed to either solver.

Rho steps and IC probes are different counted units. The strong lane counters below describe internal lane retirement and cycle handling; nonzero local capped_walks do not mean a process failed or reached the overall cap. Common calibrated operation-equivalent S stays null.

Pair	IC probes	Rho walk steps	Local capped lanes	Fruitless cycles
M1	8845441	10203425	148	1845
M2	8845441	9436758	147	1699
M3	8845441	5353957	87	973
M4	8845441	14229982	234	2510
M5	8845441	13159092	187	2319

Sources: R03, R04, R05, R07.

SECTION 10

Host, validation and admission boundaries

The host receipt describes virtualized KVM Linux x86_64, kernel 6.18.44, AMD EPYC 9V74 exposed model, five guest logical cores and one NUMA node. Cgroup RAM is 16 GiB with CPU quota 4. Each child has CPU-3 affinity and a 12 GiB virtual-address cap. Native isolation finishes exit 0 with contended_samples=0 and threads_moved=29. These are guest monitoring observations, not physical-host exclusivity or sealed ecbench L2.

All 22 children completed verified: two excluded warm-ups, ten A/A children and ten matched children. There are no process failures, overall caps or excluded pairs. Fresh independent arithmetic replay verifies all 22 answers. Final default-feature cargo test --release --locked --lib passes 3947 tests with zero failures and 112 ignored (260.41 s). Delivery runner tests pass 7/7, native exporter 4/4, and final combined example tests pass IC 10/10 plus strong 2/2. These validation checks are separate from timed execution.

The frozen baseline IC executable predates the isolated discarded-RNG-draw collector repair. The strong adapter was compiled after it, but neither measured solver calls that generic collector path. The repair is a reproducibility fix outside this measured algorithm, not an IC optimization or a reason to rewrite frozen timing receipts.

The exact legacy wide-field IC adapter is absent from the current sealed ecbench path. Canonical candidate/workload identities remain null, common S is uncalibrated, the full rank/log transcript is missing and there is no second-host timing/replay admission. Conductor invocation returned command-not-found, and no tracked .conductor/WORKFLOW.md was available. These gaps remain explicit; this report does not promote vs_rho.

- Correct the old packed-reference interpretation and wrong-modulus alias in new prose; preserve old frozen receipts.
- Next measure independent unseen targets with a prospectively frozen preparation/selector policy; five seed repeats of Q do not resolve target tails.
- For full algebraic preparation certification, retain every admitted rank relation, its witness and setup scalar, rank/matrix evidence, the solved factor logs and independent [L_j]G=R_j replays.
- For a construction-from-curve objective, newly measure original representative selection/construction; for admission, add exact ecbench integration, calibrated common costs and applicable isolation/second-host gates.
- Use event-driven child completion in a future runner, while retaining CPU totals and producer timestamps; leave this frozen panel’s wall unchanged.

Sources: R04, R07, R08, R09, R10. Validation logs: validation_logs/full_lib_tests_final.log and validation_logs/examples_tests_final.log.

SECTION 11

Evidence register and reproducible report source

This is an accounting/reference correction and a source-bound diagnostic of the existing retained candidate. It introduces no selector, algebraic identity, asymptotic improvement or admitted performance row. The quantitative plots lay out native recorded values and intervals; Python performs no statistical estimation, arithmetic replay or benchmark computation. The authoritative executed identity is records/frozen_panel_inputs.sha256, checked unchanged after the panel and corroborated by every child start/execution receipt. Earlier frozen_before_execution hashes are preparation history only and are not used as executed bindings.

The report lives beside the raw records at research/n83_evaluation_20261012/reports/run-20261012T0226Z. Native JSON copies under data/ preserve the recorded source outputs; figures.py, editable SVG diagrams, report_content.json and report_source.md reproduce its layout.

Canonical graph review found the n83 retained scoreboard anchor concerns the distinct a=0 / 81-bit engineering capacity study. The boundary ledger already labels the old a=1 observation exploratory; no controlled frontier or ledger row is promoted here. The scope of graph/source checks is recorded in the study README.

Evidence / source pin	SHA-256 or exact reference
Native primary statistics	43b00e963b379aaf718a16b864a3125a866a8be9f7f83b8eaa109d23c3d1560e
Native secondary export	b8101e76589558c8329831e3fe4649ea010661a34c6653bdc2fec40b5063b0b3
Independent fresh replay	980a3d7d52db96a559dd36ad12e317c9b12be9f1c161a2d0d86803aafcc72441
Paired-record input	e1f1b019ed5f80a739b0acc9c2e7f971d8e3400f113b47700067b0a64f8c0215
Frozen IC executable	983f28400b7852fc6628ba97d3ea83b2ab9e32d9944658c7eea831ee89a4abf2
Frozen strong executable	98949b7127baeb1fa55420d23dc13604069fcc8fde8d92b8f2e6bebd09718542
Executed native runner	8da1b77f9521fccdc5d8490a493146f0a6e2af92a2fab07c4e2e374954b9d2b4
Executed input protocol	ac7266f6caad2a4142ae06884ee748f980f9ee200c4e2649a0149f59e91adf29

Native source hashes above identify the copied recorded outputs and executables. Links below lead to the durable study records and source explanations.

ID	Evidence / path
R01	Frozen protocol: runner/protocol.json
R02	Native paired statistics: records/panel/statistics.json
R03	Native secondary export: records/panel/report_data_secondary.json
R04	Independent fresh replay: verification/fresh_replay_receipt.json
R05	Three spaces and signed-orbit witness: verification/INVARIANTS.md
R06	Curve-identity erratum: verification/CURVE_ID_ERRATUM.md
R07	Source audit: source_audit/AUDIT.md
R08	Host and isolation records: records/README.md
R09	Study delivery and validation: README.md
R10	Native runner source: runner/src/main.rs
R11	Native secondary exporter: descriptive_summary/src/main.rs

Sources: R01, R02, R03, R04, R05, R06, R07, R08, R09, R10, R11.

RESEARCH SYNTHESIS

Unifying vector bases and invariants

A cross-repository synthesis of curve families, algorithm combinations and verified performance

12 October 2026

Evidence, algorithms, and measurement protocol

Contents

A unified map of the approaches, the observed performance, and the evidence needed for the next decision.

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SECTION 01

What is fastest, and what is actually established

> Recommendation: keep rho as the qualified cold baseline; validate the degree-83 online lead and the oriented j=0 implementation on matched, admitted hosts; retain separate kernel, online and cold scoreboards.

The work fits one coherent framework: choose an exact curve and subgroup, select a representation and a usable factor base, exploit only verified subgroup actions, solve point decompositions, accumulate independent relations, and recover and replay a scalar. The best choice depends on which of these stages dominates. There is no supported universal winner across these curve families, hardware targets and timing boundaries.

For a solve from scratch, the strongest retained complete cold comparison favors signed-Frobenius rho in all six tested W64 cells at degrees 37, 41 and 53. The compact-orbit index-calculus (IC) candidates cost 1.314–22.675 times the paired rho CPU cost. These are fixed-policy results, with declared noise gates and separate correctness replay. They do not rule out other IC methods.

For one target after preparation, the degree-83 IC pipeline is an exploratory lead. Three repeats on one public point yield recorded rho/IC ratios 8.181, 5.550 and 12.671. Excluded logged rank/preparation receipts range from 199.504 to 305.046 seconds across the three runs; the ledger records R1’s 305.046-second value. Host contention, unmatched resource information and an unresolved rho setup boundary prevent a controlled speedup claim.

The clearest reusable complete-solve engineering improvement is the prime-field j=0 oriented rho walk with width-2 tau setup. Across 160 paired targets it improves recorded online wall time by median factors 2.785–3.375 against the former sixfold walk. All 320 answers independently replay. The timing remains exploratory on a heavily loaded ARM64 host.

On the RTX PRO 6000, the fastest retrieved same-function sigma update preset is fused sigma plus nibble tables at 15.884369 billion complete updates/s. The audited campaign build remains at about 14.64 billion/s; the different table walk reaches 20.08 billion/s, but its effective cost per solve has not been established. Update throughput alone cannot select the fastest attack.

Sources: R01, R02, R03, R08, R17, R18.

SECTION 02

Scope, provenance and evidence strength

This is a synthesis and audit of the available research, not a new timed cryptanalytic campaign. The three repository snapshots were inspected through connected GitHub reads on 12 October 2026. Five related PDF reports and a binary-field summary were recovered from connected Drive. Repository indexes and canonical boundary reports guided selection; the audit is broad but does not assert that every historical branch, issue or unindexed file has been exhausted.

Sources were pinned to the commits below. Historical benchmark programs retain their own earlier source freezes; a document's current commit is not automatically the measured binary's revision. The report records exact artifact paths and preserves historical labels instead of inventing current-schema candidate IDs.

During this audit, archived prime-j=0 outcomes were independently replayed with ordinary affine arithmetic: curve membership, generator subgroup order, identical target pairing, and all recovered scalar equations were checked. The published target-pair bootstrap calculation was reproduced. Degree-83 phase totals and ratios were arithmetically recomputed; cold-panel intervals were read from the sealed analysis and inverted without inventing new samples.

Four claim types are kept distinct: **derived** identities or structural bounds; **measured** outcomes on a stated workload; **independently verified** correctness or recomputation; and **proposed/model** routes or conditional estimates. A confidence interval does not establish correct accounting or hardware isolation. Missing completion costs remain unknown.

Source repository	Inspected commit	Role
aburan28/crypto	a8194d616ed8c04489ec46177d369f1b1670360a	Native IC, GPU client, factor-base search and boundary ledger
aburan28/cryptanalysis	f48351f7dbfb7535bff94e2e1d58321bb9a53404	C/Sage implementations, candidate contract, orbit synthesis and paired rho panels
aburan28/crypto-autoresearcher	7f455cfac57c6d75a6009ec84b1e13866e870c83	Evidence ledger, certified diagnostics, models and pilot history

Sources: R01, R11, R31.

SECTION 03

Five different meanings of a basis

Most apparent contradictions disappear when 'basis' is attached to its mathematical object. A normal basis for a field is not a small Frobenius-stable factor-base subspace. An orbit representative is not an ordinary unweighted relation-matrix column. A lower Macaulay degree is not a lower relation-matrix rank.

Separate semantic invariants from implementation predictors. Exact subgroup order, verified eigenvalue actions, the actual usable point set, cofactor classes and phase-aware relation equations define correctness. Register count, conversion density, product-space dimensions, solver degree and cache footprints predict cost for a particular representation and backend.

A coordinate change preserves the curve problem only through an explicit field isomorphism and point transport. Reinterpreting the same bit string under another irreducible modulus changes its field element. A genuinely changed subspace or point set is a new candidate, even if dimension and cardinality are unchanged.

A useful invariant ledger is (ℓ, dim V^2, dim V^3, dim(V·Frobenius(V)), dim(V∩Frobenius(V)), trace image, cofactor-class support, orbit-size distribution, B, K). Attach a domain and proof/measurement status to each entry. Hamming weight is coordinate-basis dependent; absolute trace and subgroup order are intrinsic. In a prime field, bit windows are encodings or constraints, not proper vector subspaces over Fp.

Object	What is fixed or measured	What it can change
Ambient field basis	Polynomial/normal/optimal normal basis; irreducible modulus; encoding and isomorphism	Multiply, square, Frobenius, conversions and canonicalization cost
Factor-base subspace V / point set D	Exact span or membership rule; usable B; set digest; trace and cofactor classes; dim V^(k)	Coverage, recognizable decompositions, solver size and useful rank
Orbit-column basis	Verified subgroup action; representative, sign and phase; effective column count K	Storage, relation count and modular coefficients
Boolean / monomial basis	Equation ideal; variable substitution; degree closure and row-space certificates	Macaulay sparsity, elimination and refutation cost
Torsion eigenbasis	E[N] basis, endomorphism action, pairing calibration and invertibility	Transported-map decoding and public leakage in isogeny constructions

Sources: R04, R05, R07, R24, R28.

SECTION 04

The unified end-to-end pipeline

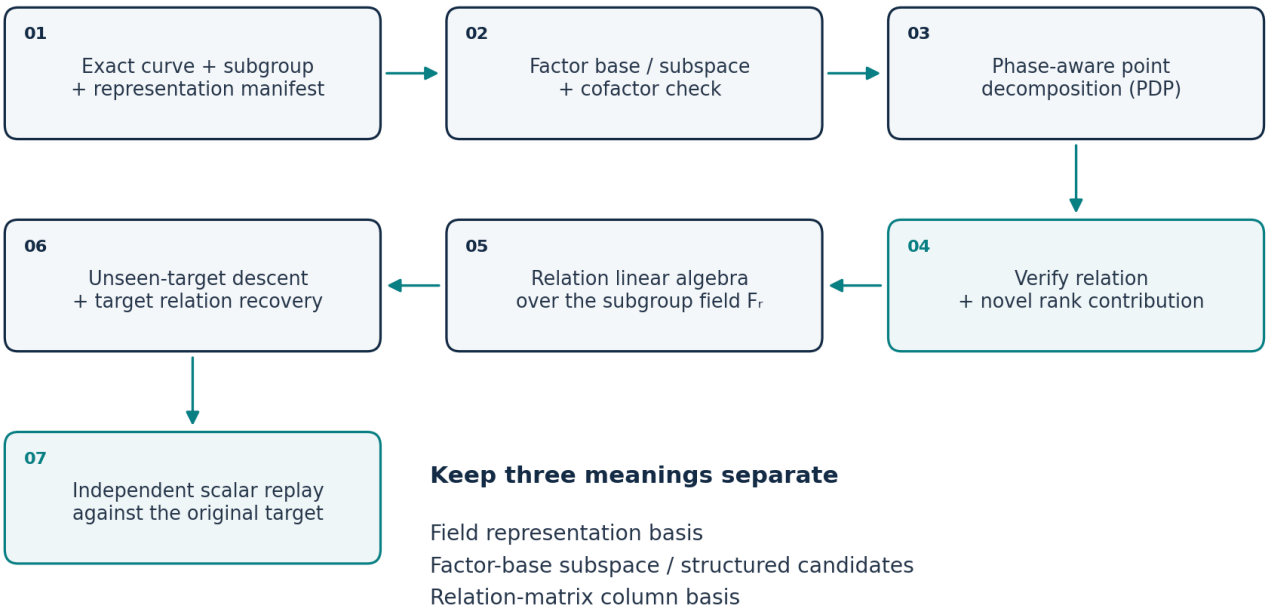
A complete candidate is the entire chain shown below. Basis choice and invariant checks belong at the points where they affect semantics or cost. Every failed query, dependent row, rejected lift and recursive descent attempt consumes resources. A partial stage record is useful evidence, but it does not occupy the complete-solve leaderboard.

Reusable preparation constructs the base, index and factor-log database without the target. Online work starts when the supplied unseen target first influences computation. Cold work includes the reusable preparation as well. The same code path may therefore lead the online scoreboard and lose from scratch.

Each admitted relation must remain a group identity after canonicalization and coefficient transport. Novel rank is measured over the subgroup scalar field; F4's internal Boolean/Macaulay elimination remains part of point decomposition. These two kinds of linear algebra have different moduli, sizes and meanings.

One measurement contract, explicit algebraic roles

Charge every attempt, failure, setup step, and final verification to its phase.



Report cold full solve and online-only solve separately; reuse is an explicit workload parameter.

A semantic and accounting diagram, not a proposed isogeny graph. Each candidate binds an exact curve/subgroup/field representation; all failures stay in the charged phase. Editable DOT and SVG are included.

Sources: R01, R11, R30.

SECTION 05

The invariants that transfer across curve families

For a prime-order subgroup G= ⟨P⟩ of order r, a verified endomorphism acts as φ(P)=[λ]P. The useful finite folding group is the subgroup generated by the actual eigenvalues in (Z/rZ)*. Abstract map names do not imply independent actions. If two maps induce the same scalar subgroup, their nominal orders cannot be multiplied.

With freely acting signed automorphism group of effective order A, generic rho receives roughly a √A reduction relative to an unfurled walk. An IC base may reduce relation columns by A, but its matched rho reference must receive the same available action. Exceptional orbits, cofactors, cycle repair and transport costs must be handled explicitly.

A high-degree endomorphism is not automatically a small finite automorphism. The tested degree-2/3 CM and Q-curve maps can have very large eigenvalue order, making small invariant bases unavailable. Conversely, an invariant union of point orbits can exist even when there is no useful small invariant linear coordinate subspace.

Family	Verified action / basis pattern	Transfer limit
Binary Koblitz E/F2	Squaring Frobenius; signed point-orbit unions; phase λ^k	Effective order and subgroup must be checked; small stable V can be absent
Prime-field j=0	(x,y)↦(ζx,y), plus sign; sixfold orbit	Available to IC and rho; oriented walk changes per-step cost
Prime-field j=1728	(x,y)↦(−x,iy), order four where rational	Field congruence and subgroup action matter
GLS / extension curves	Stable lines and Frobenius-type maps	Measured composite actions can collapse: 4 rather than 8, or 3 rather than 9
General ordinary curves / isogeny routes	Explicit maps, kernel checks and subgroup transport	No free fold from trace, conductor or a proposed neighbor
Supersingular torsion KEM	Rank-two torsion eigenbasis and pairing	Different objective; public hints can create attacks and calibration obligations

Sources: R05, R08, R24, R27.

SECTION 06

Frobenius-stable subspaces: the exact obstruction

In a normal basis, Frobenius acts as cyclic shift. Stable F2-subspaces are submodules of F2[X]/(X^n−1). For odd n, factorization of X^n−1 gives the allowed dimensions as sums of irreducible factor degrees. This is the right structural screen before testing any proposed stable vector base.

At n=131, ord_131(2)=130, so the only stable dimensions are 0, 1, 130 and 131. At n=83, ord_83(2)=82 gives 0, 1, 82 and 83. Dimension-one points are rational torsion in the relevant primary construction and disappear under its subgroup projection. These obstructions do not apply to small unions of point orbits or closures of general subspaces.

Prime degree alone is insufficient to claim that all nontrivial stable subspaces are absent. At n=31, the nontrivial factors have degree five, and useful dimensions five and six exist. The blanket prime-degree statement in the linearized-half README must be read with the multiplicative-order condition supplied by the fuller classification.

The practical search should enumerate admissible module blocks first, then choose between a stable linear base, a general subspace plus orbit union, or a point-defined orbit base. It should not sweep dimensions that cannot exist.

Field degree n	Factor-degree pattern	Achievable stable dimensions
7	1, 3, 3	0, 1, 3, 4, 6, 7
31	1 and six degree-5 blocks	0, 1, 5, 6, 10, 11, …, 31
83	1, 82	0, 1, 82, 83
127	1 and eighteen degree-7 blocks	0, 1, 7, 8, 14, 15, …, 127
131	1, 130	0, 1, 130, 131
163	1, 162	0, 1, 162, 163

Sources: R04, R10, R26.

SECTION 07

Product closure, trace and cofactor constraints

For an ℓ-dimensional subspace V in a prime-degree binary extension, the retained product-space bound is $\dim(V \cdot V) \geq \min(n, 2\ell-1)$. A scaled geometric progression $c \cdot \mathrm{span}(1, g, \ldots, g^{\wedge}(\ell-1))$ attains the unsaturated $2\ell-1$ profile when g has full extension degree and $2\ell-1 \leq n$. A nontrivial proper intermediate subfield can have $\dim(V \cdot V) = \ell$ for $1 < \ell < n$ only when ℓ divides n; F2 itself remains a proper subfield at prime degree. Normal/random-looking coordinate vectors can saturate the product space sooner and create a more expensive algebraic PDP system.

Trace explains many easy refutations. For $x3 \neq 0$ on $y^2+xy=x^3+a\,x^2+b$, write $S3=(x1x2)^2+x3\,x1x2+x3^2(x1^2+x2^2)+b$. Then $\mathrm{Tr}(S3/x3^2)=\mathrm{Tr}(x1)+\mathrm{Tr}(x2)+\mathrm{Tr}(\sqrt{b}/x3)$. If $V \subset \ker \mathrm{Tr}$, class-one targets $\mathrm{Tr}(\sqrt{b}/x3)=1$ are impossible and refute at degree two regardless of the other geometry of V.

That exact identity was checked in the retained source on 221,184 small-field evaluations. It is a screening rule and a stratification rule: compare class-zero natural targets when measuring the hard part of a trace-zero base; retain class-one rejection costs separately. A high UNSAT rate from impossible cosets is not evidence that attainable decompositions are cheap.

Cofactors impose a second filter. A target in $\langle P \rangle$ requires the summands' cofactor-torsion classes to cancel. A retained cofactor-two toy has zero odd-summand decompositions for either of two base sizes, while even-summand queries succeed. Base size alone does not imply usable yield or rank.

The source's shape-search narrative contains unsettled provenance: a frozen scale-eight run is described as pending while later conclusions cite a half-scale smoke run. Its exact identities and product bounds are usable; its empirical universal shape rankings remain provisional until the frozen tables are checked.

Sources: R04, R06, R28.

SECTION 08

Yield conservation: measure distribution and useful rank

Let D contain B distinct nonidentity points in a finite abelian group of order N. If c_D(R) counts size-m multisets summing to R, then Σ_R c_D(R)=binomial(B+m−1,m). The mean for a uniform target is therefore binomial(B+m−1,m)/N for every base of that size. For distinct unordered subsets, use binomial(B,m); typed or signed domains require their own exact tuple counts.

If the base is contained in the DLP subgroup, take N=r. If unprojected base points lie in the full curve group while targets are conditioned on the subgroup, count only cofactor-cancelling tuples. The whole-group binomial formula divided by r does not automatically describe that conditioned law.

This does not make all bases equivalent. Geometry can change the fraction of targets with any decomposition, the concentration of multiple decompositions, solver recognizability, row sparsity and rank. It cannot increase the uniform mean while keeping the tuple domain and actual B fixed.

The retained battery has exact mean deviation zero across 144 cells. A Sidon-like base has a measured coverage ratio 1.1071, while a concentrated small-multiples base has ratio 0.0021 at the largest toy size. Neither number measures the cost of finding a relation or the time to solve a logarithm.

Replace 'relations per target' as a lone selector with verified natural coverage, charged cost per accepted relation, cost per novel rank increment, time to the required rank, and target-extraction success. State the sampling law. Liftable-x sampling or known-log/short-multiple bases can bias the experiment or introduce circularity.

Metric	Use	Guard
Exact mean tuple multiplicity	Conservation sanity check	Same actual B and tuple/sign policy
Coverage Pr[c_D(R)>0]	Chance of a useful natural query	Uniform declared subgroup targets; interval and censoring
Novel rank per charged query	Value of admitted relations	Verified modular coefficients and independent rank
Cost to full rank	Reusable factor-log preparation	Failed queries, dependencies and LA charged
Verified target solve time	Selection objective	Same point and timing/resource boundary

Sources: R12, R11, R31.

SECTION 09

Phase-aware orbit compression and field representation

For a signed-Frobenius representative Ṕ, store P=ε τ^(−k)(Ṕ), not just the canonical x. On the subgroup, log(P)=ε λ^(−k) log(Ṕ) mod r. A relation's column coefficient must retain every sign and phase; contributions in the same orbit are added modulo r and may cancel.

For odd n and x≠0, Tr(y/x) supplies a Frobenius-invariant sign bit: negation sends y/x to y/x+1, so the trace flips, while squaring preserves it. The exact hypotheses matter. Composite degree and exceptional short orbits require the recorded tie and subgroup rules.

Polynomial coordinates make native carry-less multiplication and sparse reduction attractive. Normal coordinates turn Frobenius into rotation and permit least-rotation/zero-gap canonicalization. The fastest implementation often keeps a packed polynomial state and evaluates fixed linear maps, instead of fully converting the state at every step.

Phase-aware root and pair records can reduce storage dramatically, but compactness changes probe locality and the memory hierarchy. Compare equal retained memory as well as equal nominal base size. Canonical keys are skewed by minimization, so indexing by their leading bits is a poor hash.

The inspected orbit manuscript is useful for representation proofs and data flow, but its README says every end-to-end row loses to rho while also naming the degree-83 online lead. The unified interpretation is: controlled cold rows lose; the degree-83 online observation is unresolved exploratory evidence; the degree-19 6.85-fold result compares to a Gröbner pipeline, not rho.

Sources: R07, R04, R18, R30.

SECTION 10

Point-decomposition oracles and compatible combinations

A compact S3 root index, a group pair table, a half-trace linear oracle, F4/F5, SAT and FES expose different bottlenecks. An equation-count reduction does not guarantee lower conflict count, degree, memory or verified solve time. Planted SAT instances are correctness controls; they cannot substitute for ordinary-query yield.

The linearized-half method uses $e1=x1+x2$ and $e2=x1x2$. The equation $S3=e2^2+xT\ e2+xT^2\ e1^2+1=0$ gives an F2-linear system with nonzero right-hand side in (e1,e2). With a progression base, its linear system has n rows and $3\ell-1$ coordinate unknowns. Enumerate the residual affine space, split $X^2+e1\ X+e2$, check membership, lift and re-add. The intended cheap regime has $3\ell-1\le n$. Residual enumeration costs 2^{δ}, with $\delta=\ell+\dim(V\cdot V)-\operatorname{rank}(M)$; row count alone does not prove a polynomial-time oracle. The existence of a four-summand decomposition imposes a separate constraint on ℓ.

Retained toy sample counts support lower PDP growth, but the full relation budget at n=131 remains around 2^{91} in that model, approximately 30 bits beyond rho. Higher-symmetric-function rows are optimistic proposals. The implemented method is not a completed ECC2K-130 attack.

Combine only components that agree on the original equations, field representation, point domain, orbit phase and resource policy. Root/PDP verification stays against the unsimplified system. Internal F4 matrix acceleration changes PDP cost; it does not accelerate the final relation LA by definition.

Combination	Current evidence	What remains to measure
Progression + trace filter + half-trace	Exact oracle checked against brute pairs; toy sample scaling	Ordinary class-zero yield; complete rank/descent
Orbit base + compact S3 roots + guided rank	Verified factor logs and degree-83 target observations	Matched online boundary and cold selection
F4 + M4RI/RREF kernel	Stage speedups and exact-output controls	Verdict mix, degree tails and full pipeline
SAT + representative/shift selectors	Much smaller formula; some arms UNKNOWN	Natural witnesses and charged per-query cost
Gray enumeration + FES	Enumeration is part of the solver	Capacity and full original-equation verifier
One/two large-prime partials + graph LA	Exact row/filter logic in limited adapters	Wide producer wiring, cycle yield and complete cost
Isogeny + invariant base	Explicit transport prerequisites on selected routes	Map/subgroup proof, natural PDP and charged route

Sources: R01, R10, R16, R26.

SECTION 11

The latest large factor-base search: exhaustive design is not execution

> The current total-runtime winner of the degree-83 finite sweep is unknown. Its exact construction and coverage records are progress, but they do not supply a completed-solve ranking.

The degree-83 search declares a finite Cartesian design of 45,360,000 tuples: 1,050 conditional base specifications times 43,200 solver configurations. Base families include public-x, hash, Gray-prefix, polynomial/random/affine subspaces, torsion-coordinate choices and Frobenius-module blocks. Solver choices include arity, splitting, symmetry, MITM, F4, SAT, WDSat, FES, large primes, threads and modular LA.

This is exhaustive over declared axes and values. It is not exhaustive over all bases, shifts, field representations, solvers, seeds or restart schedules. Many combinations are capacity-blocked, structurally incompatible or missing an adapter. The ordinal audit accounts for those dispositions; it does not mean millions of complete solves were run.

The primary a=0 degree-83 curve uses an approximately 81-bit prime subgroup; its a=1 diagnostic has an approximately 53-bit subgroup. The exploratory online a=1 record uses yet another declared polynomial representation. These are separate workloads. Field degree alone must never merge their performance.

The completed 7,200-second pilot includes construction, verification, storage and guarded model-capacity attempts. A 2,048-column base has 339,968 points; a structured dimension-12 orbit closure has 2,001 columns and 332,166 points. The latter five-summand model has 111,008 variables and 1,120,347 clauses under its cap. Those are successful construction/capacity results.

Natural relations, rank, solver completion, complete cold runtime and the selected base remain null in the latest retained record. Two guarded m=5 trials reach their conflict cap without a relation; another trial has no solver verdict after daemon loss. Preserve these outcomes instead of selecting a 'fastest' arm from model size or construction time.

Sources: R26.

SECTION 12

One-target online evidence: the degree-83 lead

The exact legacy workload is a=1 on $y^2+xy=x^3+x^2+1$ over GF(2^83), modulus z^83+z^7+z^4+z^2+1, subgroup r=8569786107849059 (about 52.928 bits), cofactor 1128547018. It uses 600 orbit columns, 99,600 base points, a 29,878,182-entry root index and guided factor-log preparation.

All three runs solve the same frozen public point and independently replay the same scalar. The known scalar remains a validation-only sidecar. Canonical candidate_id and workload_id are null in the derived report; the legacy run IDs are retained in the artifact appendix.

The table uses the derived recorded IC online interval, with its five exclusive phases, and the derived rho recorded interval. Phase sums differ from the recorded IC interval by less than 0.001 ms, a rounding or unattributed residual retained in the audit. These are slightly different from the README's rounded process receipts. The median of paired recorded ratios is 8.1813. No interval is created from three repeats on one point: they measure that point's timing variation, not a distribution of fresh targets.

The retained rank/preparation receipts are R1 305.045905083 s, R2 199.504066584 s and R3 209.024826875 s, using 12 guided-rank workers and excluded from the online ratio. The current ledger retains R1; the README separately summarizes rank acquisition at about 4.8 minutes. Derived online receipts record IC peak RSS up to 5,811,257,344 bytes (5.811 decimal GB); memory depends on the receipt’s phase boundary. IC extraction takes 8,845,441 root probes in each run. Rho takes 5,958,775–12,179,440 walk steps. Probes and walk steps are different units. The whole-pipeline operation-equivalent cost S remains unknown.

The host is shared with unrelated work; the rho interval includes setup with unresolved target dependence; matched resource and identity manifests are incomplete. The result is a strong candidate for controlled validation, not a qualified online or cold winner.

Legacy run	Excluded prep s	IC online wall s	Rho recorded s	Recorded rho/IC	Replay
R1 (…061 / IC21)	305.046	7.939138	64.952483	8.181302	PASS
R2 (…062 / IC22)	199.504	7.746014	42.987580	5.549639	PASS
R3 (…063 / IC23)	209.025	5.623275	71.253690	12.671208	PASS

Sources: R03, R01, R30.

SECTION 13

Complete cold evidence: all six tested IC policies lose

The frozen W64 panel starts fresh processes and charges base/index construction, full-rank collection, modular solve, all public-target descents, scalar checks and artifact output. The metric is Linux wait4 child user+system CPU seconds, not online wall time. The source pairs each compact A/B geometric-mean process cost with the same-block signed-Frobenius normal-basis rho cost.

The one-target rows answer a cold one-target question. The 1,024-target rows answer a separate shared-setup batch question; their rho reference is also a batch. Do not use those batch ratios as a same-point single-target online baseline.

All 135 children finish; 90 full-rank traces and 46,170 arm-target logarithms agree with the separate replay. Prospective A/A and quiet-host gates admit all six rows under that frozen protocol. This is stronger than the older unqualified comparisons, but it does not establish physical-host exclusivity under newer evidence tiers.

Every reported 95% interval lies above parity in IC/rho cost. The closest cell, n41 with 1,024 targets and 255 columns, needs a 23.9% reduction in complete IC CPU cost merely to meet its median reference. The corresponding n53 batch needs 37.7%. Kernel gains below those complete-cost gaps are insufficient on their own.

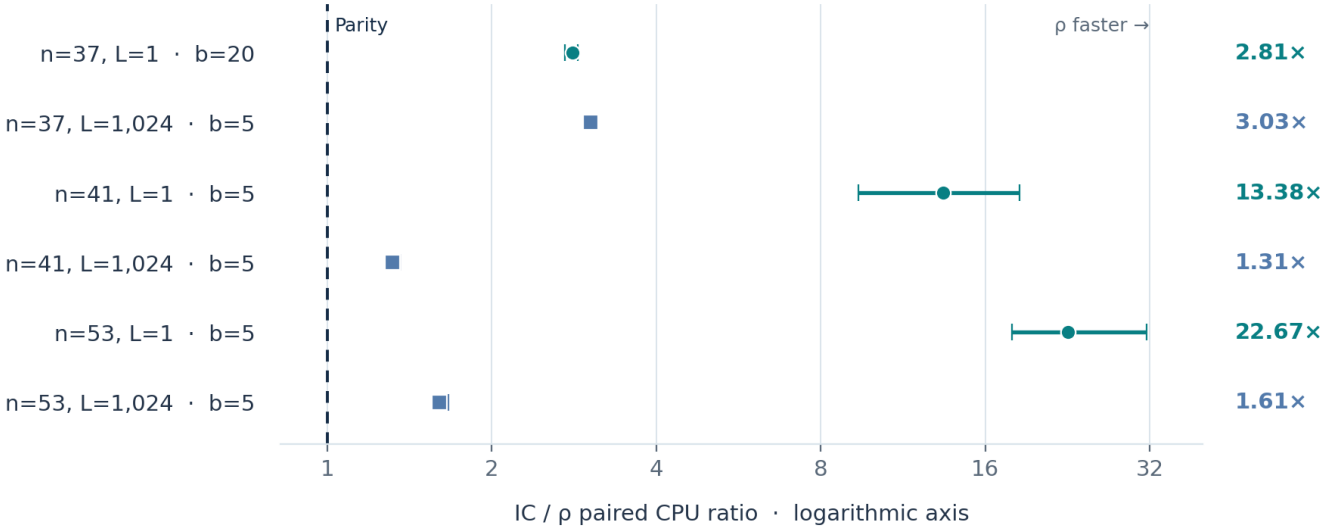
Cells run on separate Ubuntu runners and different exposed CPU models. Compare paired ratios within cells; their absolute seconds are not a cross-degree scaling fit.

A separate held-out base-size experiment illustrates the interaction: shrinking K85→64 at n41 and K220→160 at n53 saves index/matrix work but increases complete in-process cold IC cost by paired medians 1.186 and 1.636. All 12 smaller/baseline cold comparisons worsen; 24 IC/rho pairs independently replay. These two-point, unisolated macOS timings are exploratory and do not alter the W64 qualified panel. [R33]

n / targets / K	IC CPU s	Rho CPU s	IC/rho [95%]	Reduction to parity
37 / 1 / 7	0.012754	0.004555	2.810 [2.718,2.874]	64.4%
37 / 1024 / 42	0.359169	0.118599	3.031 [2.998,3.047]	67.0%
41 / 1 / 85	0.542401	0.039262	13.378 [9.367,18.491]	92.5%
41 / 1024 / 255	1.422171	1.082178	1.314 [1.288,1.359]	23.9%
53 / 1 / 220	5.717557	0.254509	22.675 [17.904,31.595]	95.6%
53 / 1024 / 440	8.802848	5.478498	1.605 [1.566,1.666]	37.7%

Cold complete-process comparison

Archived paired ratios and their source-reported 95% log-t intervals



Metric: full child-process user + system CPU; includes setup and verification. Ratios above 1 favor ρ.

Source-reported paired IC/rho CPU-cost medians and 95% log-t intervals. Values above 1 cost more than rho. n37/L1 has 20 independent paired blocks; each other cell has 5. Batches are explicitly marked. [R02]

Sources: R02.

SECTION 14

Prime-field j=0: a coherent complete-solve combination

The oriented sixfold rho walk composes with width-2 tau-adic scalar setup. Both paired builds use that same setup, so the table below isolates the walk change rather than multiplying gains from unrelated runs. Width-four variants remain separate and do not establish a uniform win.

Write ψ(x,y)=(βx,−y), acting as λ on the prime subgroup. If Y=ψ^e(C) for its orbit representative C, precompute oriented jumps ψ^e(M_i) and coefficient multiples λ^e(α_i,β_i). Use an orbit-invariant jump partition. Then F(ψ(Y))=ψ(F(Y)) on the ordinary classes, so collided classes remain merged.

The former walk canonicalizes six images and transports both coefficients each step. The oriented walk determines orientation cheaply, uses modular-addition coefficient updates, and canonicalizes the stored copy at a distinguished point. It preserves the recorded two-cycle rule and expands the jump table up to 192 entries. All jump preparation, walk attempts, collision recovery and internal scalar verification are inside the online interval.

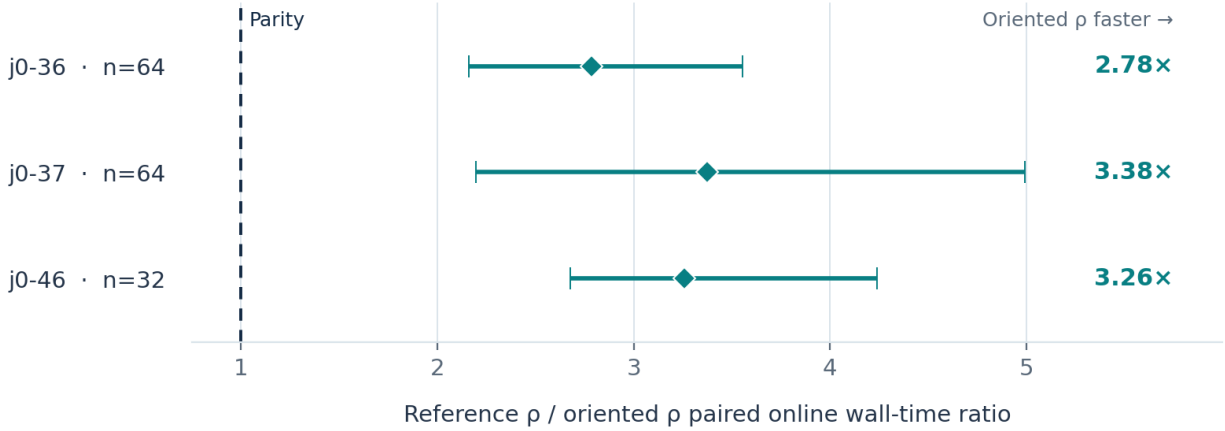
The paired panels use p=2^61−1 with b=5,11,13 and three different subgroups. The historical label j0-37 actually has a 38-bit order. There are 64,64,32 independent target pairs and no failed rows. During this audit all 320 archived solve outputs were independently replayed, and the published paired bootstrap intervals reproduced.

These timings come from a Darwin ARM64 host with load roughly 47–70 and no isolation receipt. They support a concrete exploratory implementation preference on the tested backend, not a 256-bit, GPU or cross-platform speedup. Rho's square-root exponent is unchanged.

b / subgroup r	Pairs	Reference / oriented ms	Paired ratio [95%]
5 / 51131959441	64	18.703 / 6.920	2.785 [2.159,3.554]
11 / 157632877033	64	52.819 / 16.458	3.375 [2.197,4.992]
13 / 42111239174233	32	762.544 / 232.414	3.260 [2.677,4.240]

Prime j=0: exploratory oriented ρ

Source medians and 95% target-pair bootstrap intervals · contended ARM64 host



n = independent target pairs; each pair has independent affine scalar replay.
Exploratory evidence: host contention and no isolation receipt limit the speed claim.

Independently recomputed archived target-pair ratios and source bootstrap intervals, n=64/64/32 pairs. Ratio is former sixfold rho online wall / oriented rho online wall. Contended ARM64 timing remains exploratory. [R08]

Sources: R08, R09.

SECTION 15

ECC2K-130 hardware: select effective progress, not headline rate

The GPU work improves implementation cost while keeping the audited sigma walk's mathematical work unchanged. Packed five-word state removes the original bitsliced spill bottleneck. Frobenius permutation networks, polynomial-state arithmetic, native carry-less products, batch depth, compact state, fused scheduling and nibble maps address successive machine bottlenecks.

The latest same-function sigma-table preset measures 15.884369 billion complete scalar updates/s against a session-local fused control at 15.513653 billion/s. Five alternating A/B pairs give median 1.023930, minimum 1.023803; five A/A pairs show maximum drift 0.099%. It passes the declared engineering threshold. The sorted 1,709,477-record validation corpora are identical; host/device arithmetic, restart and checkpoint tests pass. DP34 collection and witness-carrying rates for this nibble-table preset are not established by the retrieved measurements.

The historical audited campaign preset measures 14.637530 billion updates/s, with DP34 collection at 14.106673 billion/s. Later witness-carrying fused mode has a different workload: 11.800903 against 9.347867 billion/s, paired median 1.262291. Do not substitute counter-free rates for witness or collection rates.

The different r-adding table walk reaches 20.08 billion/s. Its as-built cycle behavior was priced at 4.85–6.26 times the sigma cost per solve; revised repair rules have projected costs but lack complete matching measurements. A faster update can generate less useful collision progress.

For selection, measure effective useful steps to a verified collision, dead/fruitless-cycle fraction, report loss, restart/probe overhead, host ingestion and collision solve under the actual collection rule. When the iteration function changes, correctness equivalence to the old distinguished-point corpus is no longer the appropriate sole gate: both correct solves and the walk constant must be measured.

Architecture changes the winner. The retained relative logic price per carry-less multiply is about 37.8 on RTX PRO 6000, 3.78 on B200 and 2.0 on H100. Each device needs its own measured full workload and representation choice; a static instruction reduction is not a portable speedup.

Preset / workload	Recorded rate, billion updates/s	What can be selected
Audited campaign sigma	14.637530; DP34 14.106673	Established campaign implementation
Fused sigma, counter-free	15.436677 vs15.015004; 5 pairs	Verified same-function engineering
Fused sigma + nibble tables	15.884369 vs15.513653; 5 pairs	Fastest retrieved same-function update preset
Fused witness-carrying	11.800903 vs9.347867; 5 pairs	Separate witness workload
Different table walk	20.08	Rate lead; effective solve-cost winner unresolved

Sources: R17, R18, R19, R32.

SECTION 16

A useful negative: sparse field bases can move the bottleneck

The sparse-basis compound screen tests z^131+z^8+z^3+z²+1 and z^131+z^8+z^5+z²+1. Both fields, basis isomorphisms and linear maps are exact. The source reducer drops from 180 to 82 word operations; those are expression-level counts, not compiled SASS instructions.

Keeping the state sparse makes the required normal/Frobenius maps dense. The proposed table route needs 275 shared-load instructions per update and 78,464 bytes per block including reservation. Transferring the recorded random shared-load rate gives a favorable lookup-only projection 15.283375 billion/s, below the verified fused reference 15.436677, even with the rest of the kernel free.

The native independent audit reproduces the exact result bytes and checks all 17,161 basis product pairs per candidate, linear map directions and dense guards. It does not launch a GPU or perform a solve. The admission gate rejects the whole-walk prototype under that frozen transfer model.

The exact eight-entry wide-load layout might behave better than the random byte-load probe; its ideal-rate sensitivity passes the gate. Therefore this is a conditional negative screen, not a hardware impossibility theorem. A small frozen layout microprobe is the justified prerequisite.

This is the general lesson for combinations: reduction, conversion, Frobenius, canonicalization, state storage and occupancy must be priced together. Selecting the sparsest modulus or fewest source operations alone can select a slower end-to-end representation.

Sources: R20.

SECTION 17

GLV, GLS, extension-field and isogeny results

The unified folding driver closes a seed set under verified maps and transports coefficients to orbit columns. Across prime, quadratic and cubic extension groups, the useful column count follows the actual eigenvalue subgroup. The measured composite action is 12 on one j=0 GLS twist, four rather than eight on the j=1728 twist, and three rather than nine for the cited j=0 cubic-subfield action.

Folding can reduce the relation count without changing rank accumulation per column. Against the same-action rho, much of the apparent advantage disappears. In the cited prime-field pilot, folded IC remains 18–2,834 times an A=2 rho reference; matched larger-action rho is cheaper still.

On line bases with $r \approx p^2$, rho costs $\Theta(p)$ and roughly $\Theta(p)$ relations are required. The retained correction moves an earlier below-rho exponent claim to a fit about 0.53. The FGHR two-torsion/conic oracle and base fold compound materially, but the best measured arm remains about 267 times rho. This is useful stage engineering, not an asymptotic break.

E18's k=4 linear-algebra model changes an asymptotic matched-rho ratio and extrapolated handover. Those are conditional model calculations, not measured solves at the proposed large sizes. Keep all order, cofactor, yield and LA assumptions with the numbers.

The volcano-position test's retained n=19 attempt ratio is 0.996 with 95% interval 0.976–1.017: a null. Conductor/level information is not a decomposition oracle. An isogeny route needs explicit maps, an ordered route, generator/target transport and kernel/subgroup checks before ISO1 can be assigned.

The prime-field integrated IC suite's initially impressive 7–18-fold ratios use an independent-target unfurled rho reference. Its fairer folded shared-DP batch model gives rho/IC around 0.72–0.89 at 28–40 bits. Those algorithm counts also omit scalar-certification cost. They cannot become a fully charged single-target headline.

Sources: R05, R27, R29.

SECTION 18

Solinas primes: arithmetic improvements with a bounded claim

The seven-term search examines p=2^n+Σ s_i 2^(e_i)+s0, signs ±1, distinct interior exponents divisible by 32, through seven total terms. It finds 1,151 proved primes across 192,224,256,384,521 bits. NIST remains the only three-term choice in the 192/224 families; no two-,three-,four-term primes occur in the declared 256/384 grid.

The lowest written term count is not the lowest reduction cost. The seven-term 256-bit word32 winner has L1 reduction score 32 versus 55 for P-256, but confirmed throughput improves only 9.6%. Different reducers rank the same primes differently; a large gain against a slow folding implementation need not beat generic remainder.

Confirmation fixes the screened pairs, uses a new seed, longer batches and 21 shuffled rounds. The intervals below are source percentile-bootstrap intervals for those selected fixed comparisons on one EPYC cloud instance. They describe multiplication plus reduction; conversions, operand generation and allocation are outside the timed region.

The portable word32 kernels have data-dependent normalization and are not a comparison of fully tuned constant-time production assembly. Changing p creates a new curve whose order, cofactor, twist and embedding-degree checks must be completed before an ECDLP comparison.

Comparison in portable word32	NIST / candidate ns	Paired throughput ratio [95%]
256-bit seven-term winner	91.1 / 83.1	1.096 [1.093,1.103]
384-bit bounded winner	137.7 / 132.3	1.043 [1.033,1.049]
192-bit nearest-prime control	56.5 / 51.9	1.078 [1.071,1.093]
224-bit nearest-prime control	66.4 / 59.8	1.115 [1.105,1.117]
256-bit nearest-prime control	91.1 / 68.9	1.319 [1.303,1.332]
384-bit nearest-prime control	137.7 / 114.9	1.195 [1.190,1.201]

Sources: R21.

SECTION 19

Number fields and point patterns: invariants are not attack budgets

Three fields must remain distinct: the coordinate finite field Fp; a characteristic-zero representation field Kf=Q(α) from a sparse polynomial; and the ordinary curve's endomorphism algebra KE=Q(π)=Q(√(t²−4p)). A single p can admit several unrelated Kf choices. Their degree, signature and discriminant do not define the elliptic subgroup's logarithm difficulty.

Example: P-192 admits X³−X−1 at radix 2^64 and X^6−X²−1 at radix 2^32, with different representation fields while the NIST curve stays fixed. The Frobenius-order discriminant t²−4p is not automatically the fundamental field discriminant; conductor and maximal order require separate computation.

The sparse-prime report establishes differences in multiplicative finite-field structure. P-224's p−1 has a smooth factorization, but its elliptic group has a different large prime order. A short multiplicative orbit of 2 at P-521 is not a short elliptic point-doubling orbit. NFS norm relations solve a multiplicative-group problem; x coordinates are not a group homomorphism that transfers generic ECDLP.

The point-pattern report's negation relation and coordinate bit-flip patterns reflect encoding and the equation. Its 16,384-point diagnostics give useful tests of the declared sampler and visible biases; an absence of significant patterns is not a hardness certificate. A sampled constant bit beyond an exact guaranteed prefix is not an exact law.

Use these studies to select arithmetic backends, verify samplers and identify structural prerequisites for a transfer. Do not attach attack-speed or security-bit claims to representation discriminants, scatterplots, term counts or isolated bit frequencies.

Sources: R22, R23.

SECTION 20

Autoresearcher: certified diagnostics and conditional frontiers

The evidence ledger provides useful falsification tools even when it establishes no complete DLP improvement. Degree-four mutant closure at n19,m2,ℓ10 refutes 400/400 UNSAT cases with Clopper–Pearson 95% [0.9908,1]; the comparison M4 refutes 0/400 with [0,0.0092]. There are 0/300 SAT false refutations, with independently checked certificates. The source explicitly makes no cost claim and reports sota_delta=0.

Fresh m=3 pilots refute 60/60,40/40,8/8 UNSAT cases at n9,n11,n15. Sparse-support null controls are 11/11 versus dense null 0/12; the n19 row is memory-censored. These are exploratory diagnostics outside the established evidence lifecycle.

Representative-plus-shift Frobenius SAT encodings reduce formulas, but retained conflict ratios do not show a universal benefit: 1.017–1.467 for one solver and 1.000–1.212 for another. Count clauses, conflicts, status mix, witness validity and complete charged time separately.

The ANF ledger estimates positive cost margins at several degrees, including +40.8 bits at n131, and a negative margin −4.9 at n409. It assumes ANF materialization, a particular mD^2 sparse LA model and monomial charges. Those conditional estimates are not proofs that a curve is protected or measured large-size attacks.

Coherent time×memory accounting shifts a modeled crossover to n460 sparse / n520 dense, without changing an exponent. EXP-IC-001's estimated finite amortization crossover in 4/13 seeds also cannot be promoted: its Gröbner calls return trivial ideals with no decompositions. A computed crossover between incomplete pipelines is not a solve.

Result class	Admissible use	Promotion still needed
Certified low-degree refutation	Correctness / algebraic obstruction	Charged natural-query mix and full solve
Pilot support/closure result	Hypothesis selection	Lifecycle evidence, independent controls and uncensored rung
ANF or time×memory model	Conditional budget screen	Feasible oracle, measured constants and complete accounting
Finite amortization estimate	Workload planning	Successful relation/rank/descent in both compared methods

Sources: R13, R14, R15, R16, R31.

SECTION 21

Torsion eigenbases: an adjacent, versioned line of work

The Eigen-Séta material uses a rank-two torsion basis aligned with a hidden endomorphism. It belongs in the basis taxonomy but not in the ECDLP-speed leaderboard. Its correctness depends on rational torsion, independent eigenlines, scalar admissibility, product-isogeny transport and torsion calibration.

The 11 October PDF proposes a three-scalar decoding construction with diagonal masks commuting with the hidden map. Its public-input experiments recover 18 recorded toy challenges by seed search using a pairing fingerprint. Ninety-six wrong discriminant guesses finish both quotient chains but fail torsion calibration. Those observations concern a specific toy distribution and provide no validated full-size parameter set.

The current repository README describes a distinct v2 secret-degree KEM/PKE prototype with nominal category parameter sets and descriptive operation times. It is not the same artifact/version as the PDF's preliminary native crate. Do not combine the PDF's evidence or security assumptions with v2's correctness timings.

The general transferable principle is that a public invariant can be both a computational aid and a leakage channel. A basis-dependent fingerprint may filter guesses; existence of a quotient/isomorphism is weaker than calibrated action on the designated torsion. Keep the exact published distribution and the verified action in the candidate record.

Sources: R24, R25.

SECTION 22

Measurement changes: separate objectives, charge exclusive phases

The user's end-to-end question needs two primary views in this report. **Cold one-target cost** answers fastest from scratch. **Online one-target cost** answers fastest after reusable preparation, matching the repository's current default IC contract. A third, explicitly secondary batch view answers amortization. None can replace another without changing the question.

Use five exclusive online phases: T_online,1=T_target_query+T_target_PDP+T_target_relation_check+T_target_descent+T_target_recovery_check. Stop after a recovered scalar is verified. Fixture generation and process launch are outside this interval. If an independent extra check is outside, record its timing separately.

Cold accounting adds target-independent setup, isogeny transport, base materialization, index/precomputation, relation-query/PDP/check work, matrix construction, relation LA and all target phases. Each timer has one owner. Nested collector time cannot be summed again with its contained solve/check timers. Missing costs are null, not zero.

Keep wall seconds, process CPU seconds, GPU-active seconds, memory peaks and I/O separately. A throughput benchmark excludes different work than a complete process. Common operation-equivalent S=charged_operations/√r is allowed only after freezing a defensible unit conversion; an S3-root probe and a rho step are not interchangeable.

For k targets, report T_total(k)=T_prepare+Σ_i T_online,i and its resource policy. A deterministic break-even model k>(P_IC−P_ref)/(t_ref−t_IC) is valid only with comparable preparation, a positive per-target difference, a declared target law and stable costs. Use distributions rather than this scalar approximation when tails or failures matter.

Prepared IC may use a large reusable log table. Retain a supplementary equal-preparation-memory rho/BSGS reference when the question is total capability under a shared budget. That control is absent from the degree-83 lead; its absence limits the claim, not the observed answer correctness.

Sources: R11, R26, R30.

SECTION 23

Measurement changes: a statistical experiment that survives selection

Freeze the mathematical instance, target distribution, candidate manifests, resource envelope, timing starts/stops and acceptance rule before measurement. Tune on a separate corpus; confirm a fixed shortlist on disjoint targets and, where meaningful, disjoint signed-Frobenius orbits. A large parameter grid requires selection-aware reporting.

Pair baseline and candidate on each identical public target. Randomize or alternate execution order within target and session; insert A/A controls. Repeats on one target characterize timing noise. Fresh independent targets characterize solve variability. Treat host/session and target as clustering units in resampling; do not count every correlated launch as an independent target.

Use within-pair log cost ratios or paired medians with intervals chosen before inspection. A reported ratio of marginal medians is a different estimator. The cold source uses a preregistered log-t interval; the j=0 source uses target-pair bootstrap. This report preserves those estimators and does not invent confidence for the degree-83 one-point sample.

Store every failure, cap, OOM and producer error. Successful-only medians select the easy subset. Publish completion probability by budget and memory cap, conditional successful time, and a censored time-to-verified-solve summary or restricted mean at a fixed budget. If more than half are censored, the unconditional median is not observed.

For relation yield, use binomial intervals only when the trials fit the independence assumptions. Correlated walks require block or cluster treatment. For rank, plot accepted rows and novel pivots against cumulative charged cost, including failures. Degree and FFD/DoR distributions include natural SAT, refuted and timeout cells separately.

An admitted CPU speedup needs the appropriate host/instance receipt, same placement/resource limits, source/workload hashes, repeated pairs and a95% interval clearing parity and the measured noise floor. A/A drift detects some noise; it does not prove absent short contention, thermal changes or physical-host exclusivity.

Report every finalist and relevant rejected combination. Do not multiply gains from different occupancy, compiler, base, target or device runs. A combined candidate needs its own paired measurement.

Sources: R02, R08, R11, R31.

SECTION 24

A coherent implementation and experiment plan

First build a single candidate/result registry with exact curve UID, subgroup order, field modulus/encoding, generator, base and point-set hashes, effective orbit columns, phase convention, PDP backend, collector, final LA, descent, source hashes and resource policy. Preserve legacy IDs and append corrected records rather than rewriting evidence.

Second repair accounting and the baselines. Recalculate older degree-41/53 overlapping intervals from retained receipts if sufficient; leave unresolved costs null. Freeze rho's target-dependent setup boundary at degree 83. Keep matched signed-Frobenius and oriented j=0 references, verified scalar replay and fair same-point timing.

Third validate the two concrete online leads: the retained degree-83 compact-root/guided-log pipeline and prime-field j=0 oriented rho with width-two tau. Run both on admitted matched hosts with independent fresh targets, A/A controls and complete exclusive phases. Do not rent or run large campaigns before the manifests and gates are executable.

Fourth pursue the cold objective with equal-useful-size base selection, starting from the measured n41/K255 and n53/K440 batch gaps, while retaining one-target cold rows as the main from-scratch view. Screen natural coverage, useful rank/cost, memory and failures; fully solve and independently verify finalists.

Fifth test interaction explicitly. Freeze a small factorial or staged ablation panel: progression versus random V; trace class; orbit fold; oracle choice; collector; row solver; field state; conversion/canonicalization backend. Record the combined candidate, not a synthetic sum of component winners. GPU batch depth and shared-memory geometry are interaction variables, not portable constants.

Finally make each new claim self-contained: exact identities and run IDs, included phases, raw rows, hardware tier, correctness certificate, intervals, selector/holdout split and failed arms. Update canonical graphs only when a verified fact or correction changes them. This audit creates new comparison visuals; it does not modify the repositories' graphs or promote a performance row.

Priority	Concrete deliverable	Exit criterion
1 · Accounting	Versioned online/cold/batch schema and receipt repair	Exclusive sums; missing costs remain null
2 · Validation	Admitted degree83 and j=0 paired target panels	Verified solves, matched resources, qualified intervals
3 · Selection	Equal-useful-size cold finalist panel	Complete cost beats matched reference on holdouts
4 · Interaction	Compatible combined candidates and ablations	Measured whole candidate, no synthetic compounding
5 · Frontier	Cheap-oracle route that survives yield and rank	Complete measured solve; model predictions stay labeled

Sources: R01, R02, R26, R30, R31.

SECTION 25

Open questions and final research judgment

> The main change to the research approach is measurement discipline: rank complete candidates on comparable tasks, preserve preparation and failures, and choose by verified completion cost. The mathematical search should use the exact invariants to eliminate impossible or circular cells before spending compute.

The most valuable unresolved question is not which invariant looks strongest in isolation. It is whether an affordable ordinary-query oracle, on a sufficiently large and useful base, survives full-rank collection and one-target extraction under an honest baseline. The known invariants mostly reorganize existing work or rule out unproductive parameter regions.

The degree-83 online lead warrants validation, but the 81-bit primary subgroup sweep is a different problem from the 53-bit diagnostic and must establish its own natural relations and complete cost. The degree 131 stable-subspace obstruction means a direct transfer from convenient small-degree module bases is unavailable; nonlinear point sets or orbit unions need new oracle evidence.

Prime-field oriented rho provides a concrete implementation stack with complete verified solves and replicated arithmetic, but its wall-time gain still needs controlled hardware confirmation. The same-action rho reference should improve whenever the IC representation work makes that improvement available.

GPU representation engineering has already yielded substantial throughput gains. The next meaningful comparison is effective cost per verified solve under cycle-safe collection and actual report/witness handling. The different table walk’s 20.08-billion update rate cannot settle that question.

Overall, unify the implementations around typed mathematical objects and shared evidence contracts, while preserving workload-specific algorithms. The retrieved evidence supports engineering advances, exact obstructions, certified diagnostics and promising online observations. It does not yet establish one fastest new complete algorithm across the portfolio.

Sources: R01, R02, R03, R05, R08, R17, R26.

SECTION 26

Artifact identities, replay and audit boundaries

The report package contains the PDF, editable structured source, Markdown source, plotting/drawing source, comparison data, retrieved text extracts, source indexes and the independent archived j=0 replay receipt. It contains no new benchmark run or promoted claim.

The degree 83 legacy IDs are N83A1K600W12We202610061R21R1, N83A1K600W12We202610062R22R2 and N83A1K600W12We202610063R23R3. Its canonical candidate and workload IDs remain null. The exact public point is Q=(265589834553716618053296 0,388152630799167326529 2615).

The cold W64 source freeze is 38c173072e16f580237447c93b085634275ba048; input-freeze SHA-256 da958a3f1117dd1b88703fed2055c5c9f64255516e1e920f35d193bf53c6a88d; sealed evidence manifest SHA-256 350779eb61b5c9b14f065f4f2b3563166de0aaa6ed1da91f3d5308a8fc6fac97. The separate correctness replay did not retime the cells.

The oriented-rho audit's exact p=2305843009213693951, b, r and generator coordinates are retained in archived_j0_validation.json beside the three raw target panels. It verifies 320 archived answers and reproduces the published 10,000-resample paired bootstrap. The result is a new correctness/statistical replay of old measurements, not a new timing run.

Canonical graph/report sources reviewed include the frobenius-orbits representation/pipeline/benchmark sections, the current boundary ledger, the volcano experiment README, the degree 83 sweep diagram/status, the GPU ledger and sparse-basis screen. No repository graph is rewritten because this synthesis does not establish a new timed frontier. Contradictory prose and missing receipt fields are called out in the report.

The original 39-prime Solinas comparison is superseded for arithmetic rankings by the seven-term/native confirmation report; the newer number-field and point-pattern reports answer separate structural questions. The Drive binary-field summary’s 14.64 headline is contextualized with the newer 15.88 same-function repository preset. Eigen-Séta PDF/v2 evidence remains version-separated.

Sources: R02, R03, R07, R18, R21, R24, R25, R26.

SECTION 27

Source register 1 of 3

Links below identify the exact inspected documents. Source-local benchmark freezes and underlying receipt links remain authoritative for the measured binary. Local archived extracts and a SHA-256 manifest accompany this report.

R01 · Current pipeline frontier and acceptance gates
crypto:docs/ic/BOUNDARY_TARGETS.md

R02 · Six-cell complete cold comparison
crypto:research/notes/ecc2k130/disjoint_cold_v2_outcome_20261001/RESULT.md

R03 · Degree-83 derived phase report
crypto:experiments/koblitz-single-target-n83-20261006/derived_exploratory_claim_report.json

R04 · Koblitz factor bases, oracles and representation history
crypto:research/notes/ecc2k130/RESEARCH_KOBLITZ_INDEX_CALCULUS.md

R05 · Endomorphism-invariant bases across families, E1–E18
crypto:research/notes/index-calculus/RESEARCH_GLV_INVARIANT_FACTOR_BASES.md

R06 · Shape, trace and product-profile analysis
crypto:research/notes/index-calculus/RESEARCH_FACTOR_BASE_SHAPE_SEARCH.md

R07 · Signed-Frobenius synthesis draft and limitations
cryptanalysis:papers/frobenius-orbits/README.md

R08 · Oriented sixfold rho: complete paired solves
cryptanalysis:experiments/prime-j0-rho-coordinate-canon-20261002/README.md

R09 · Tau setup and width comparisons
cryptanalysis:experiments/prime-j0-tau-20260930/README.md

R10 · Linearized-half oracle and whole-pipeline budget
cryptanalysis:experiments/linearized-half-decomposition/README.md

R11 · Candidate measurement contract
cryptanalysis:experiments/ic-candidate-catalog/MEASUREMENT.md

SECTION 28

Source register 2 of 3

Links below identify the exact inspected documents. Source-local benchmark freezes and underlying receipt links remain authoritative for the measured binary. Local archived extracts and a SHA-256 manifest accompany this report.

R12 · Exact decomposition-yield conservation
crypto-autoresearcher:knowledge/findings/KN-FIND-007.md

R13 · Independently certified degree-4 closure diagnostic
crypto-autoresearcher:knowledge/findings/KN-FIND-c3a917.md

R14 · ANF cost-model evidence
crypto-autoresearcher:ledger/evidence/EV-ICPERF-784b25.yaml

R15 · Coherent time–memory model accounting
crypto-autoresearcher:ledger/evidence/EV-SEMBIN-4125ec.yaml

R16 · Latest binary-field pilots and censored outcomes
crypto-autoresearcher:analysis/ecdlp2m-iteration-20261009/README.md

R17 · ECC2K-130 hardware synthesis and caveats
crypto:paper/ecc2k130/README.md

R18 · Fused sigma plus nibble-table measured preset
crypto:ecc2k130/SIGMA-TABLE.md

R19 · Walk quality, cycles and cost per solve
crypto:ecc2k130/WALK-CONSTANT.md

R20 · Sparse-basis compound negative static screen
crypto:ecc2k130/benchmarks/sparse-basis-compound/RESULTS.md

R21 · Solinas primes through seven terms
Connected Drive PDF

R22 · Solinas primes: number-field effects
Connected Drive PDF

SECTION 29

Source register 3 of 3

Links below identify the exact inspected documents. Source-local benchmark freezes and underlying receipt links remain authoritative for the measured binary. Local archived extracts and a SHA-256 manifest accompany this report.

R23 · Solinas point-pattern diagnostics
Connected Drive PDF

R24 · Eigen-Séta proposal dated 11 October
Connected Drive PDF

R25 · Eigen-Séta v2 secret-degree prototype
crypto:eigen_seta/README.md

R26 · Degree-83 finite design and exact pilot completion status
crypto:research/koblitz_n83_factor_base_sweep_20261008/README.md

R27 · Volcano-position hypothesis and null comparison
cryptanalysis:experiments/volcano-ic/README.md

R28 · Product-subspace and PDP degree profiles
cryptanalysis:experiments/pdp-degree-heuristics/README.md

R29 · Prime-field orbit-base pipeline and corrected rho comparison
cryptanalysis:suite/docs/ic/README.md

R30 · Phase overlap, counters and claim-check repairs
crypto:docs/ic/PLAN_IC_ACCOUNTING_FIXES_20261007.md

R31 · Evidence and quantitative-claim policy
crypto-autoresearcher:docs/progress-fidelity.md

R32 · Fused sigma measurements and witness workload
crypto:ecc2k130/benchmarks/sigma-fused/RESULTS.md

R33 · Held-out base shrink raises complete rank cost
crypto:experiments/koblitz-base-size-cold-panel-20261004/RESULT.md